



SKAVIA

Solution-Oriented Service & Workforce Platform

SOFTWARE REQUIREMENTS SPECIFICATION (SRS).

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements of SKAVIA, a solution-oriented service and workforce platform. The purpose of this document is to describe, in a formal and unambiguous manner, what the SKAVIA system is intended to do so that developers, database designers, user-interface designers, system analysts, testers, and academic reviewers share a single, consistent understanding of the system to be built.

This document translates the SKAVIA project idea, as described in the source project documents, into a structured requirements-engineering artifact. It does not introduce technologies, algorithms, or business rules that are not supported by the source material; where an implementation detail has not yet been decided, this document explicitly states that the detail is "to be finalized during implementation."

1.2 Scope

SKAVIA is a platform that connects people and businesses who have a problem or requirement with the skilled workers needed to solve it. Unlike a conventional marketplace, in which a customer must independently search for and select a worker, SKAVIA is designed to receive a description of a requirement, understand it, identify the skills involved, find and rank suitable workers, form a team when the requirement demands multiple skills, support coordinated execution of the resulting work, and manage payment, completion, and reputation.

The scope of the system described in this SRS includes the following major areas:

- Account creation, authentication, and role/mode-based access control.
- Maintenance of Worker, Individual Customer, Business, and Organization profiles under a unified account model.
- Category, service, and skill management, and search/discovery of workers and services.
- Submission of simple and complex requirements by clients.
- Requirement analysis to identify category, skill(s), and structured details from a requirement description.
- Rule-based worker matching and ranking.
- Individual service booking, acceptance, execution, and completion.
- Team formation, Team Leader assignment, and project/task management for complex requirements.
- In-app communication and notifications.
- Payment processing, worker earnings tracking, and fee calculation.
- Reviews, ratings, and worker reputation management.
- Worker verification and trust management.
- Complaint and dispute handling.
- Administration across Super Admin, Company Admin, and Branch Manager roles, including branch-based specialization.
- Reporting and analytics for administrators and business clients.

Advanced capabilities that are described in the source material as future or exploratory concepts — such as machine-learning-based matching, an autonomous AI Team Builder, automated price/duration estimation, and platform-wide expansion — are documented separately in Section 6.2, "Future Enhancements," and are not treated as guaranteed current functionality. This SRS describes SKAVIA as a single, coherent system and does not divide the system into development phases.

1.3 Objectives

- To formally specify the functional behavior expected of the SKAVIA platform for all defined user roles.
- To specify the non-functional qualities (performance, security, usability, reliability, and others) that the system must exhibit.
- To model the system through use case, entity-relationship, sequence, and activity diagrams that are consistent with the written requirements.
- To define a database structure capable of supporting all specified functionality.
- To provide a requirement traceability matrix connecting requirements to use cases, modules, database entities, and diagrams.
- To clearly separate currently specified functionality from future enhancement concepts.

1.4 Target Users

SKAVIA is intended to be used by the following categories of people and organizations:

- Individual Customers who need a single service performed (e.g., a household repair).
- Business Clients who need one or more services, potentially forming a complete business solution (e.g., launching a restaurant).
- Organizations with recurring or larger-scale service needs.
- Workers / Skill Providers who offer one or more skills for individual or team-based engagements.
- Team Leaders, who are workers temporarily assigned coordination responsibility for a specific project.
- Branch Managers, who oversee a specialization branch of the platform.
- Company Admins, who manage SKAVIA's day-to-day internal operations.
- The Super Admin, who governs the platform as a whole.

1.5 Definitions, Acronyms, and Abbreviations

Term	Definition
SKAVIA	The solution-oriented service and workforce platform described in this document.
Requirement	A structured or natural-language description of a problem or need submitted by a Customer, Business Client, or Organization.
Simple Requirement	A requirement that can be fulfilled by a single worker with a single relevant skill.
Complex Requirement	A requirement that requires multiple skills and, therefore, a team of workers.
Matching Engine	The rule-based component that scores and ranks candidate workers against a structured requirement.

Term	Definition
Team Leader	A worker assigned coordination and reporting responsibility for a specific team/project.
Branch	A specialization unit within SKAVIA that focuses on a particular category of services (e.g., SKAVIA Home).
Reputation	An aggregated indicator of a worker's reliability and quality derived from completed work and reviews.
Verification	The process by which SKAVIA confirms a worker's identity, skill, or certificate claims.
Dispute	A formally escalated complaint that requires investigation and a recorded resolution decision.
SKAVIA Service/ Management Fee	The portion of the client's total payment retained by SKAVIA for operating the platform and coordinating the service.
Mode	The active context (Client Mode or Worker Mode) that a dual-role user is operating in within a single account.

2. Overall Description

2.1 Product Perspective

SKAVIA is a new, self-contained platform rather than an extension of an existing system. It is best understood as a solution-oriented service and workforce platform: a client does not need to know which specific worker, or how many workers, they require. The client describes a problem or requirement, and SKAVIA is responsible for understanding the requirement, identifying the skills involved, finding suitable workers, forming a team where necessary, assigning coordination responsibility, managing execution, supporting communication, handling payment, confirming completion, and collecting ratings and reviews that build worker reputation over time.

This distinguishes SKAVIA from a conventional freelancer or service marketplace, in which the burden of searching, comparing, and selecting a worker rests entirely on the customer. In SKAVIA, that burden is shared: the customer expresses a need in their own words (a simple need such as an electrical repair, or a complex need such as launching a restaurant), and the platform's Requirement Engine and Matching Engine actively assist in resolving that need into the correct worker or team.

SKAVIA is organized around specialization branches (for example, SKAVIA Home, SKAVIA Technical, SKAVIA Business) that allow categories of services to be managed with domain-appropriate oversight, while all branches operate under a single overall SKAVIA ecosystem, account model, and administrative structure. Branches are an organizational and operational concept within one system; they are not separate applications.

2.2 Product Functions

At a high level, SKAVIA performs the following functions, summarized here and specified in full detail in Section 3.1:

- Account, authentication, and profile management for all user types.
- Category, service, and skill catalog management, and search/discovery.
- Requirement submission and requirement analysis (structuring free-text requirements into actionable data).
- Rule-based worker matching and ranking.

- Individual service request life-cycle management (request, accept/reject, execute, complete).
- Team formation, Team Leader assignment, and project/task management for complex, multi-skill requirements.
- In-app communication and role-appropriate notifications.
- Payment processing with transparent separation of worker payment and SKAVIA's service/management fee.
- Review, rating, and reputation management.
- Worker verification and trust indicators.
- Complaint submission and dispute resolution.
- Multi-level administration (Super Admin, Company Admin, Branch Manager) with branch-based specialization.

2.3 User Classes and Characteristics

SKAVIA defines eight user classes. A single physical person may hold more than one profile type under one unified account (for example, a person may be both a Worker and an Individual Customer), but the internal SKAVIA administration roles (Super Admin, Company Admin, Branch Manager) are organizationally distinct from external client roles (Individual Customer, Business Client, Organization) and are never combined with them.

Role	Core Responsibility	Permission Level
Super Admin	Full platform governance: manages Company Admin accounts, branches, platform-wide settings, security settings, categories/skills taxonomy, and global reports. Does not typically perform day-to-day operational matching or booking actions.	Highest — platform-wide, unrestricted within governance scope
Company Admin	SKAVIA's internal operational management: manages workers, customers, businesses, verification, complaints, disputes, payments, and reviews platform reports. Internal staff role, distinct from any external client.	High — operational, platform-wide
Branch Manager	Manages a specific specialization branch (e.g., SKAVIA Home, SKAVIA Technical): oversees branch-level workers, requirements, teams, and projects, and participates in verification and complaint handling within the branch.	Medium — scoped to assigned branch
Team Leader	A Worker temporarily assigned coordination responsibility for a specific team/project: assigns tasks, monitors progress, communicates with the client, and confirms completion.	Medium — scoped to assigned project/team
Worker / Skill Provider	Registered service provider: manages profile/skills/availability, receives and responds to service or team requests, executes work, and receives payment and reviews.	Standard — scoped to own profile and assigned work
Individual Customer	External end-user seeking a single service: submits requirements, books workers, communicates, pays, and reviews.	Standard — scoped to own requirements and bookings
Business Client	External business seeking one or more services or a complete business solution: submits requirements, receives proposed teams, manages projects, pays, and reviews.	Standard — scoped to own requirements and projects
Organization	External organizational entity with requirements similar to a Business Client, potentially recurring or larger scale.	Standard — scoped to own requirements and projects

A Company Admin is an internal SKAVIA operations role responsible for day-to-day platform management (verification, complaints, payments oversight, worker/business administration). A Business Client is an external customer using SKAVIA to obtain services, workers, teams, or complete business solutions. These two roles are never interchangeable: a Company Admin does not submit requirements as a customer, and a Business Client has no administrative authority over other users' data. This distinction is preserved consistently across the use case model, functional requirements, database model, and sequence diagrams presented in Section 4.

2.4 Operating Environment

SKAVIA is delivered as a single cross-platform mobile application built with Flutter and the Dart programming language, targeting current versions of Android and iOS from one shared codebase. The application communicates with back-end services provided by Supabase, which supplies the primary PostgreSQL-compatible relational database, auto-generated data access APIs, and real time capabilities described throughout Section 3 and Section 4. User authentication, session management, and account verification are handled through Supabase Auth. Worker portfolio images, certificates, requirement attachments, and other media files are stored using Firebase Storage. Development is carried out using Android Studio and Visual Studio Code as the primary integrated development environments. Any implementation detail not covered by this confirmed stack (for example, the payment gateway or push-notification delivery mechanism) is to be finalized during implementation rather than assumed.

2.5 Design and Implementation Constraints

- All access to protected functions must be governed by role-based and mode-based access control, consistent with the role model in Section 2.3, and enforced through the application's authorization logic together with Supabase row-level security policies.
- Personal, financial, and verification data must be protected from unauthorized access, consistent with the non-functional requirements in Section 3.2.
- Core workflows (requirement submission, matching, booking, project management, payment) depend on network connectivity between the Flutter mobile application and Supabase/Firebase back-end services.
- Where SKAVIA relies on an AI-assisted advisory component for requirement analysis, the system must remain operable through a manual, form-based fallback if that component is unavailable, since the AI component assists rather than replaces core business logic.
- The payment mechanism(s) supported must allow the client's total payment to be transparently divided into a worker payment component and a SKAVIA service/management fee component.
- The architecture must be modular enough to accommodate additional branches, categories, services, and skills without structural redesign.
- Development resources and timeline are limited, as reflected in the realistic academic timeline in Section 5.2.

2.6 Assumptions and Dependencies

- It is assumed that users provide accurate profile, requirement, and availability information; the system's matching quality depends on this accuracy.
- It is assumed that workers maintain reasonably up-to-date availability information.
- The ability to fulfill a requirement depends on the presence of suitably skilled, verified, and available workers already registered on the platform.
- Payment processing depends on the availability and correct integration of the payment mechanism(s) selected during implementation.
- Notification delivery depends on the availability of underlying communication channels (in-app, and any additional push-notification channel added during implementation) and network connectivity.
- Worker verification depends on the accuracy of information submitted by the worker and on timely administrative review.
- It is assumed that SKAVIA operations staff are available to review and confirm team composition and Team Leader assignment for complex requirements, since this SRS does not treat automated team formation as guaranteed current functionality (see Section 6.2).

3. Specific Requirements

3.1 Functional Requirements

Functional requirements are grouped by module and identified with unique IDs. Each requirement is written using the mandatory form "The system shall..." and is assigned a priority (High, Medium, or Low) reflecting its importance to the core SKAVIA workflow described in Section 1.2. Requirement groups A through U correspond directly to the modules referenced throughout this document and in the Requirement Traceability Matrix (Section 4.8).

3.1.A Authentication and Account Management

ID	Requirement	Priority
FR-A01	The system shall allow a new user to register using an email address or phone number, a password, and a selected primary role.	High
FR-A02	The system shall verify a new account through an email or phone verification code before granting full access.	High
FR-A03	The system shall authenticate registered users through a login mechanism before allowing access to protected functions.	High
FR-A04	The system shall allow an authenticated user to log out and invalidate the active session/token.	Medium
FR-A05	The system shall allow a user to reset a forgotten password through a verified email or phone channel.	High
FR-A06	The system shall allow an authenticated user to change their account password from account settings.	Medium
FR-A07	The system shall allow an administrator to activate or deactivate a user account.	High
FR-A08	The system shall enforce role-based and mode-based authorization so that a user may only access functions permitted by their account, role, active mode, and permission level.	High

ID	Requirement	Priority
FR-A09	The system shall maintain a single unified account per person that may hold a Worker Profile, a Customer Profile, and/or a Business/Organization association simultaneously.	High
FR-A10	The system shall allow a user holding both a Worker Profile and a Customer Profile to switch between Client Mode and Worker Mode without creating a separate account.	Medium
FR-A11	The system shall record an audit log entry for significant administrative account actions such as activation, deactivation, and role changes.	Medium

3.1.B User and Profile Management

ID	Requirement	Priority
FR-B01	The system shall allow a user to create and update basic personal profile information, including name, photo, and contact details.	High
FR-B02	The system shall allow a Worker to create and maintain a skill-based Worker Profile containing skills, experience, location, working area, availability, expected rate, portfolio, and certificates.	High
FR-B03	The system shall allow an Individual Customer to create and maintain a Customer Profile containing contact information, location, and preferences.	High
FR-B04	The system shall allow a Business Client to create and maintain a Business Profile containing business name, business type, contact information, location, and description.	High
FR-B05	The system shall allow an Organization to create and maintain an Organization Profile containing organization name, organization type, and contact details.	Medium
FR-B06	The system shall allow a Worker to add, edit, and remove portfolio items demonstrating previous work.	Medium
FR-B07	The system shall allow a Worker to add and manage certificates associated with their profile.	Medium
FR-B08	The system shall allow a Worker to set and update availability information for defined dates or periods.	High
FR-B09	The system shall allow a user to view their own booking, service, and project history.	Medium
FR-B10	The system shall allow any user to view another user's public Worker Profile, including skills, ratings, reviews, and verification status.	High

3.1.C Category and Service Management

ID	Requirement	Priority
FR-C01	The system shall allow an authorized administrator to create, edit, and disable a service category.	High
FR-C02	The system shall allow an authorized administrator to create, edit, and disable a service under a category.	High
FR-C03	The system shall allow an authorized administrator to create a skill and associate it with one or more services.	High
FR-C04	The system shall present categories and their associated services in a browsable hierarchical structure to users.	Medium
FR-C05	The system shall allow a Worker to associate their profile with one or more existing skills.	High

3.1.D Search

ID	Requirement	Priority
FR-D01	The system shall allow a user to search for workers or services by skill.	High
FR-D02	The system shall allow a user to search for workers or services by category.	High
FR-D03	The system shall allow a user to search for workers or services by location or working area.	High
FR-D04	The system shall allow a user to filter search results by availability, rating, verification status, and expected rate.	Medium
FR-D05	The system shall present search results in a ranked, paginated list.	Medium

3.1.E Requirement Submission

ID	Requirement	Priority
FR-E01	The system shall allow a Customer, Business Client, or Organization to submit a requirement containing a title, description, category, location, required date, and optional budget and attachments.	High
FR-E02	The system shall assign a unique identifier and an initial status to every submitted requirement.	High
FR-E03	The system shall allow a requirement to be edited by its submitter prior to the completion of matching.	Medium
FR-E04	The system shall allow a submitter to attach supporting files to a requirement.	Low
FR-E05	The system shall allow a submitter to view the history and current status of their submitted requirements.	Medium

3.1.F Requirement Analysis

ID	Requirement	Priority
FR-F01	The system shall analyze a natural-language requirement description to identify the applicable category and candidate required skills.	High
FR-F02	The system shall determine whether a requirement is a simple, single-worker requirement or a complex, multi-skill requirement.	High
FR-F03	The system shall request clarifying information from the submitter when essential information such as location, date, or required skill is missing.	Medium
FR-F04	The system shall convert an analyzed requirement into a structured requirement record usable by the Matching Engine.	High
FR-F05	The system shall provide a manual, form-based method of structuring a requirement when the AI-assisted analysis service is temporarily unavailable.	High
FR-F06	The system shall not fabricate skills, experience, certificates, or qualifications when structuring requirement or profile text.	High

3.1.G Worker Matching

ID	Requirement	Priority
FR-G01	The system shall calculate a match score for each candidate worker using a rule-based weighted scoring model based on skill match, location, availability, experience, rating, verification status, and expected rate.	High
FR-G02	The system shall return a ranked list of candidate workers for each required skill in a structured requirement.	High
FR-G03	The system shall exclude candidate workers who do not meet mandatory constraints of a requirement, such as being unavailable during the requested period.	High
FR-G04	The system shall provide a human-readable explanation of a worker's ranking based on existing profile and requirement data.	Low
FR-G05	The system shall allow authorized SKAVIA operations personnel to review and adjust the candidate selection before a team is confirmed for a complex requirement.	High

3.1.H Individual Services

ID	Requirement	Priority
FR-H01	The system shall allow a Customer to request or book a matched worker for a single-worker requirement.	High
FR-H02	The system shall allow a Worker to accept or reject an incoming service request.	High
FR-H03	The system shall track the status of a service request through defined states: Requested, Accepted, Rejected, In Progress, Completed, and Canceled.	High
FR-H04	The system shall allow a Worker to mark an in-progress service as completed.	High
FR-H05	The system shall require Customer confirmation of completion before the related payment is finalized.	High

3.1.I Team Management

ID	Requirement	Priority
FR-I01	The system shall allow authorized SKAVIA operations personnel to create a team composed of matched candidate workers for a complex requirement.	High
FR-I02	The system shall allow workers to be added to or removed from a team.	Medium
FR-I03	The system shall allow a specific responsibility or role to be assigned to each team member.	Medium
FR-I04	The system shall track team status through defined states: Forming, Active, and Completed.	Medium
FR-I05	The system shall provide a team-internal communication channel for team members.	Medium
FR-I06	The system shall derive overall team progress from the status of the tasks assigned to its members.	Medium

3.1.J Team Leader

ID	Requirement	Priority
FR-J01	The system shall allow authorized SKAVIA operations personnel to assign one worker as Team Leader for a given team and project.	High
FR-J02	The system shall allow a Team Leader to view the details of their assigned project and team composition.	High
FR-J03	The system shall allow a Team Leader to create and assign tasks to team members.	High
FR-J04	The system shall allow a Team Leader to monitor task and overall project progress.	High
FR-J05	The system shall allow a Team Leader to submit progress updates that are visible to the client.	Medium
FR-J06	The system shall allow a Team Leader to report issues to SKAVIA administration.	Medium
FR-J07	The system shall allow a Team Leader to confirm overall work completion prior to client sign-off.	High

3.1.K Project Management

ID	Requirement	Priority
FR-K01	The system shall create a Project record when a requirement is resolved through multi-worker or team-based execution.	High
FR-K02	The system shall store, for each Project, the originating requirement, client, assigned team, deadlines, and current status.	High
FR-K03	The system shall track Project status through the defined sequence: Requested, Planning, Team Formed, In Progress, Review, and Completed.	High
FR-K04	The system shall allow a client to view the progress and current status of their Project.	High
FR-K05	The system shall allow files and documents to be shared within the context of a Project.	Medium
FR-K06	The system shall require client confirmation before a Project is marked as Completed.	High

3.1.L Task Management

ID	Requirement	Priority
FR-L01	The system shall allow a Team Leader to create a task with a title, description, assigned team member, and deadline.	High
FR-L02	The system shall allow a Worker to view tasks assigned to them.	High
FR-L03	The system shall allow a Worker to update task progress and mark an assigned task as complete.	High
FR-L04	The system shall allow a client to review submitted work associated with a task and request a revision where applicable.	Medium
FR-L05	The system shall track task status through defined states: Assigned, In Progress, Submitted, and Completed.	High

3.1.M Communication

ID	Requirement	Priority
FR-M01	The system shall provide in-app messaging between a client and a worker associated with an active service request.	High
FR-M02	The system shall provide in-app messaging between a client/business and the assigned Team Leader of a project.	High
FR-M03	The system shall provide a team-internal communication channel among members of the same team.	Medium
FR-M04	The system shall restrict communication access to users who share a defined role-based relationship to the relevant service or project.	High
FR-M05	The system shall allow an administrator to communicate with users for support or complaint-handling purposes.	Medium

3.1.N Notifications

ID	Requirement	Priority
FR-N01	The system shall notify a Worker of new job or service opportunities matching their skill profile.	Medium
FR-N02	The system shall notify relevant parties when a service request is accepted or rejected.	High
FR-N03	The system shall notify relevant parties of project and task status updates.	Medium
FR-N04	The system shall notify relevant parties of payment-related events.	High
FR-N05	The system shall notify a Worker of changes to their verification status.	Medium
FR-N06	The system shall notify involved parties of complaint and dispute status changes.	Medium
FR-N07	The system shall notify relevant parties of upcoming deadlines or scheduled bookings.	Low

3.1.O Payment

ID	Requirement	Priority
FR-O01	The system shall calculate the total client payment as the sum of the worker payment and the SKAVIA service/management fee, and shall keep both amounts separately identifiable.	High
FR-O02	The system shall record and track payment status through defined states: Pending, Completed, Failed, and Refunded.	High
FR-O03	The system shall support one or more payment methods; the specific payment gateway(s) and merchant integration are to be finalized during implementation.	Medium
FR-O04	The system shall generate a receipt or invoice record for each completed payment.	Medium
FR-O05	The system shall support cancellation and refund handling in accordance with defined payment status transitions.	Medium

3.1.P Worker Earnings

ID	Requirement	Priority
FR-P01	The system shall record worker earnings separately from the SKAVIA fee for every completed payment.	High
FR-P02	The system shall allow a Worker to view total, pending, and available earnings.	High
FR-P03	The system shall allow a Worker to view an itemized earnings history per service or project.	Medium
FR-P04	The system shall allow an administrator to monitor aggregate worker earnings and platform fee revenue.	Medium

3.1.Q Reviews and Reputation

ID	Requirement	Priority
FR-Q01	The system shall allow a client to submit a rating and a written review after confirming completion of a service or project.	High
FR-Q02	The system shall calculate and update a Worker's reputation score based on ratings and completed work history.	High
FR-Q03	The system shall display a Worker's aggregated rating, review count, and reputation indicator on their profile.	High
FR-Q04	The system shall prevent submission of a review prior to confirmed completion of the associated service or project.	Medium

3.1.R Verification

ID	Requirement	Priority
FR-R01	The system shall allow a Worker to submit verification information relevant to their skill category, such as identity documents, certificates, or evidence of experience.	High
FR-R02	The system shall allow an authorized administrator to review a verification submission and approve it, reject it, or request additional information.	High
FR-R03	The system shall display a verification indicator on the profile of an approved Worker.	Medium
FR-R04	The system shall track verification status independently for each verification type submitted by a Worker.	Medium

3.1.S Complaints and Disputes

ID	Requirement	Priority
FR-S01	The system shall allow a user to submit a complaint referencing a specific service request or project, together with a category, description, and optional attachments.	High
FR-S02	The system shall track complaint status through defined states: Open, Under Review, Assigned, Resolved, and Closed.	High
FR-S03	The system shall allow an authorized administrator to escalate a complaint into a formal dispute.	Medium
FR-S04	The system shall allow an authorized administrator to record a dispute resolution decision and close the associated case.	Medium
FR-S05	The system shall notify the complainant of status changes and the final resolution of their complaint or dispute.	Medium

3.1.T Administration

ID	Requirement	Priority
FR-T01	The system shall allow the Super Admin to manage platform-wide settings, security settings, and Company Admin accounts.	High
FR-T02	The system shall allow a Company Admin to manage workers, customers, businesses, verification, complaints, disputes, and payments at the operational level.	High
FR-T03	The system shall allow a Branch Manager to manage branch-level workers, requirements, teams, and projects.	High
FR-T04	The system shall allow an authorized administrator to manage categories, services, and skills.	High
FR-T05	The system shall allow an authorized administrator to activate or deactivate user accounts.	High
FR-T06	The system shall allow an authorized administrator to view platform-wide reports and analytics.	Medium
FR-T07	The system shall maintain an audit log of significant administrative actions for accountability purposes.	High

3.1.U Reporting and Analytics

ID	Requirement	Priority
FR-U01	The administration dashboard shall display key operational metrics, including total users, active workers, active projects, revenue, pending verification, and open complaints.	Medium
FR-U02	The system shall allow an administrator to view reports filtered by branch, category, or date range.	Medium
FR-U03	The system shall allow a Business Client to view reports related to their own projects and payments.	Medium

3.2 Non-Functional Requirements

Non-functional requirements describe the qualities the system must exhibit rather than specific behaviors. Measurable targets are provided where they can be reasonably specified at this stage; where a precise numeric target depends on infrastructure decisions not yet made, this document states that the target is to be finalized during implementation, rather than asserting an unrealistic guarantee.

Performance

ID	Requirement	Priority
NFR-P01	The system shall return search and matching results within a reasonable response time under normal operating load; specific target thresholds are to be finalized during implementation and performance testing.	High
NFR-P02	The system shall be designed to handle concurrent requests from multiple users without service degradation under expected operating conditions.	Medium

Security

ID	Requirement	Priority
NFR-S01	The system shall authenticate users through Supabase Auth before granting access to any protected resource or function.	High
NFR-S02	The system shall enforce role-based access control for administrative, operational, and financial functions, implemented through the application's authorization logic and Supabase's row-level security policies.	High
NFR-S03	The system shall rely on Supabase Auth's credential handling so that passwords are never stored or transmitted in plain text by the SKAVIA application.	High
NFR-S04	The system shall protect payment and personally identifiable information in transit and at rest using the security mechanisms provided by Supabase and Firebase Storage; any additional application-level protection is to be finalized during implementation.	High

Usability

ID	Requirement	Priority
NFR-U01	The system shall present role-appropriate dashboards so that each user type is shown only the functions relevant to their role and permissions.	Medium
NFR-U02	The system shall provide clear feedback messages for successful actions, validation errors, and system errors.	Medium

Reliability

ID	Requirement	Priority
NFR-R01	The system shall preserve data consistency for financial and status-related transactions such as payments, bookings, and task updates.	High
NFR-R02	The system shall provide a graceful degraded mode (e.g., manual requirement structuring) when an AI-assisted advisory component is unavailable, rather than blocking the core workflow.	Medium

Availability

ID	Requirement	Priority
NFR-A01	The system shall be designed for high operational availability appropriate to a production service platform; specific availability targets are to be finalized during implementation.	Medium

Maintainability

ID	Requirement	Priority
NFR-M01	The system shall be developed using a modular architecture that separates the requirement engine, matching engine, project/team management, payment, and administration modules to support independent maintenance and future enhancement.	Medium

Scalability

ID	Requirement	Priority
NFR-SC01	The system shall be designed to allow the addition of new categories, services, skills, and branches without requiring structural redesign.	Medium
NFR-SC02	The system architecture shall support horizontal scaling of core services as the number of users and transactions grows; the specific scaling approach is to be finalized during implementation.	Low

Data Integrity

ID	Requirement	Priority
NFR-D01	The system shall enforce referential integrity between related entities such as requirements, matches, bookings, teams, projects, and payments.	High
NFR-D02	The system shall validate mandatory fields and data formats before persisting user-submitted data.	Medium

Privacy

ID	Requirement	Priority
NFR-PR01	The system shall restrict access to personal, financial, and verification data to authorized roles only.	High
NFR-PR02	The system shall handle user data in accordance with applicable data protection principles; specific regulatory compliance measures are to be finalized during implementation.	Medium

Compatibility

ID	Requirement	Priority
NFR-C01	The system's application interface shall be delivered as a single cross-platform mobile application built with Flutter, supporting current Android and iOS versions; specific minimum OS versions are to be finalized during implementation.	Medium

3.3 External Interface Requirements

A. User Interface

- The system shall provide a role-appropriate dashboard within the SKAVIA Flutter mobile application for each of the eight defined user classes, showing only the modules and actions relevant to that role and current mode (see Section 4.9 for the module structure by role).
- The system shall provide interfaces for requirement submission that accept free-text description input, with optional structured fields (category, location, date, budget).
- The system shall present search and matching results in a clear, ranked, and filterable list.
- The system shall display the SKAVIA logo consistently across login/signup, navigation, dashboards, and notification surfaces, preserving its original proportions and color identity.

B. Software Interfaces

- The Flutter mobile application shall communicate with Supabase through its auto-generated data APIs and official client SDK for all data read/write operations described in Sections 3.1 and 4.4.
- The Flutter mobile application shall communicate with Supabase Auth for account registration, login, session/token management, and verification.
- The Flutter mobile application shall communicate with Firebase Storage for uploading and retrieving worker portfolio images, certificates, and requirement/complaint attachments.
- The system shall interface with a payment processing mechanism to complete client payments and worker payouts; the specific gateway is to be finalized during implementation.
- Where an AI-assisted requirement analysis component is used, the Requirement Engine shall interface with it through a defined internal service boundary, treating it strictly as an advisory input rather than a source of final business decisions.

C. Hardware Interfaces

- The system does not require specialized hardware beyond a standard Android or iOS mobile device capable of running the Flutter application and a network connection.

D. Communication Interfaces

- The system shall communicate over standard internet protocols (e.g., HTTPS) between the Flutter mobile application and the Supabase/Firebase backend services.
- The system shall support delivery of notifications through in-app channels; any additional push-notification channel is to be finalized during implementation.

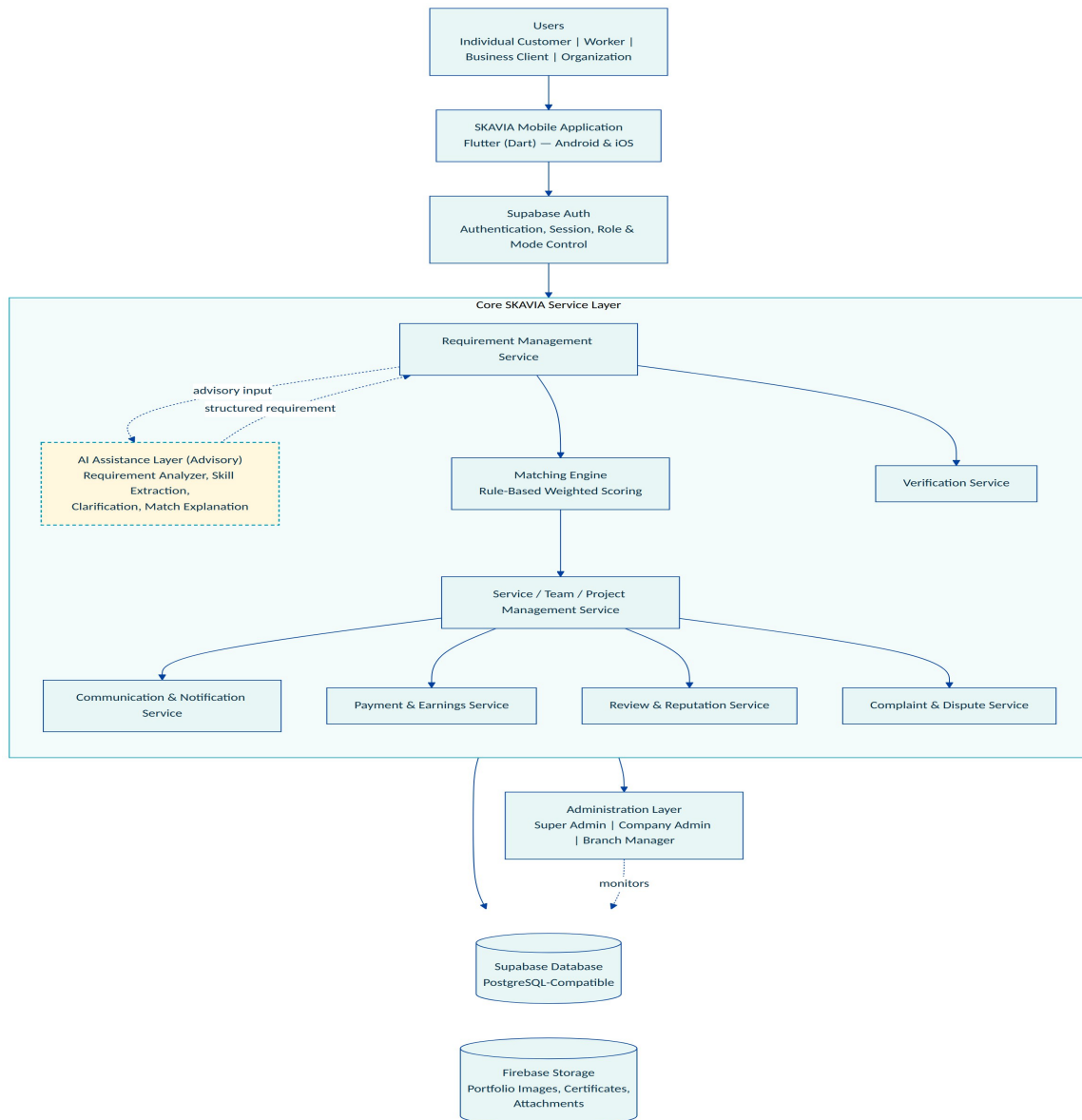
4. System Models

This section presents the system models that translate the requirements in Section 3 into concrete structural and behavioral diagrams. All diagrams were produced directly from the requirements and role model defined in this document and are internally consistent with one another, with the database structure in Section 4.4, and with the functional requirements in Section 3.1. This section is laid out in landscape orientation to accommodate the width of the diagrams.

4.1 System Overview / Architecture Diagram

The following diagram shows the logical architecture of SKAVIA. Users interact with the system through the Flutter mobile application, which enforces authentication (via Supabase Auth) and authorization before requests reach the core service layer. The core service layer contains the Requirement Management Service, an AI Assistance Layer used strictly in an advisory capacity, the rule-based Matching Engine, and the Service/Team/Project Management, Communication, Payment, Review, and Verification services, all operating against Supabase as the primary data and API layer. Administration operates above the core service layer with governance and monitoring access. Files and images (portfolios, certificates, attachments) are persisted separately in Firebase Storage, while all structured/relational data is persisted in the Supabase (PostgreSQL-compatible) database.

Figure 4.1 — SKAVIA System Architecture



The AI Assistance Layer (dashed box) provides advisory input to the Requirement Management Service only; it does not independently execute core business logic such as final worker selection, payment, or administrative decisions.

4.2 Use Case Diagrams

Because SKAVIA has a wide actor and use case base, the use case model is divided into logically consistent diagrams: (1) Individual Service use cases for Individual Customers and Workers, (2) Business Solution and Project use cases for Business Clients, Organizations, Team Leaders, and Workers, and (3) Administration use cases for Super Admin, Company Admin, and Branch Manager, further split into Parts A and B to keep each diagram readable. Each diagram remains consistent with the complete system and with the functional requirements in Section 3.1.

4.2.1 Use Case Diagram — Individual Service

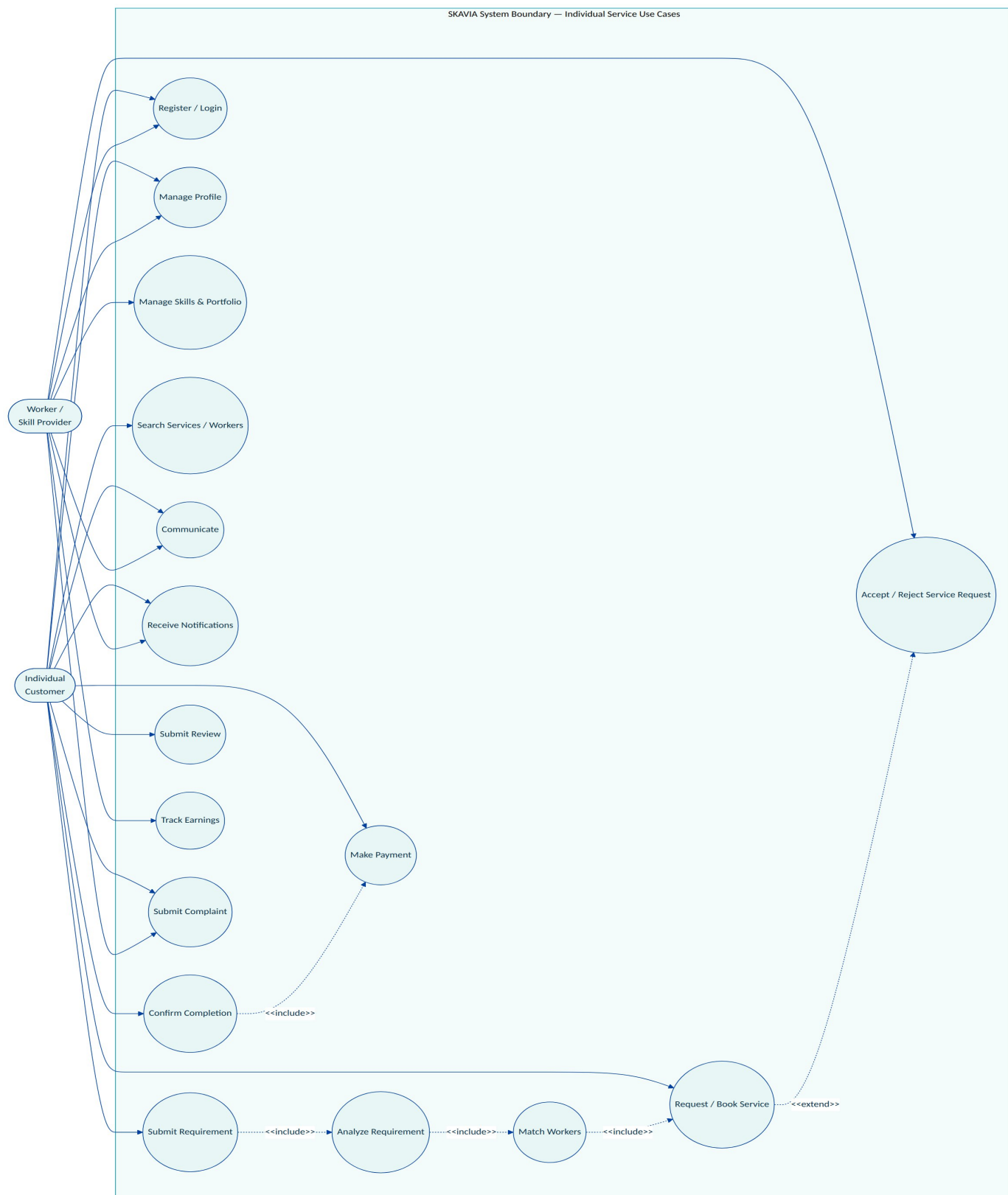
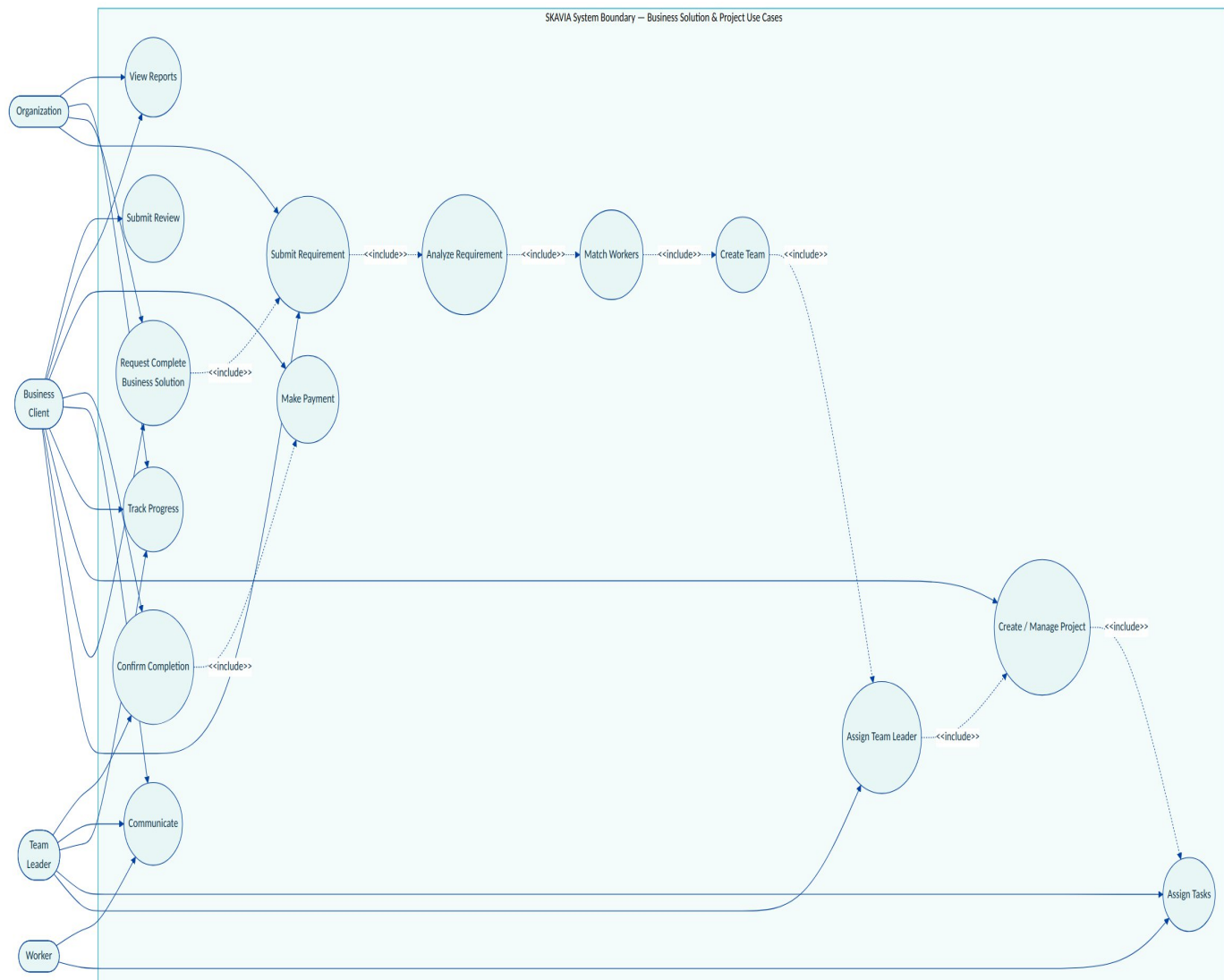


Figure 4.2 — Customer / Worker Use Case Diagram

Submitting a requirement includes requirement analysis, which includes worker matching, which includes the request/booking use case; accepting/rejecting a request extends the booking use case.

4.2.2 Use Case Diagram — Business Solution and Project

Figure 4.3 — Business / Project Use Case Diagram

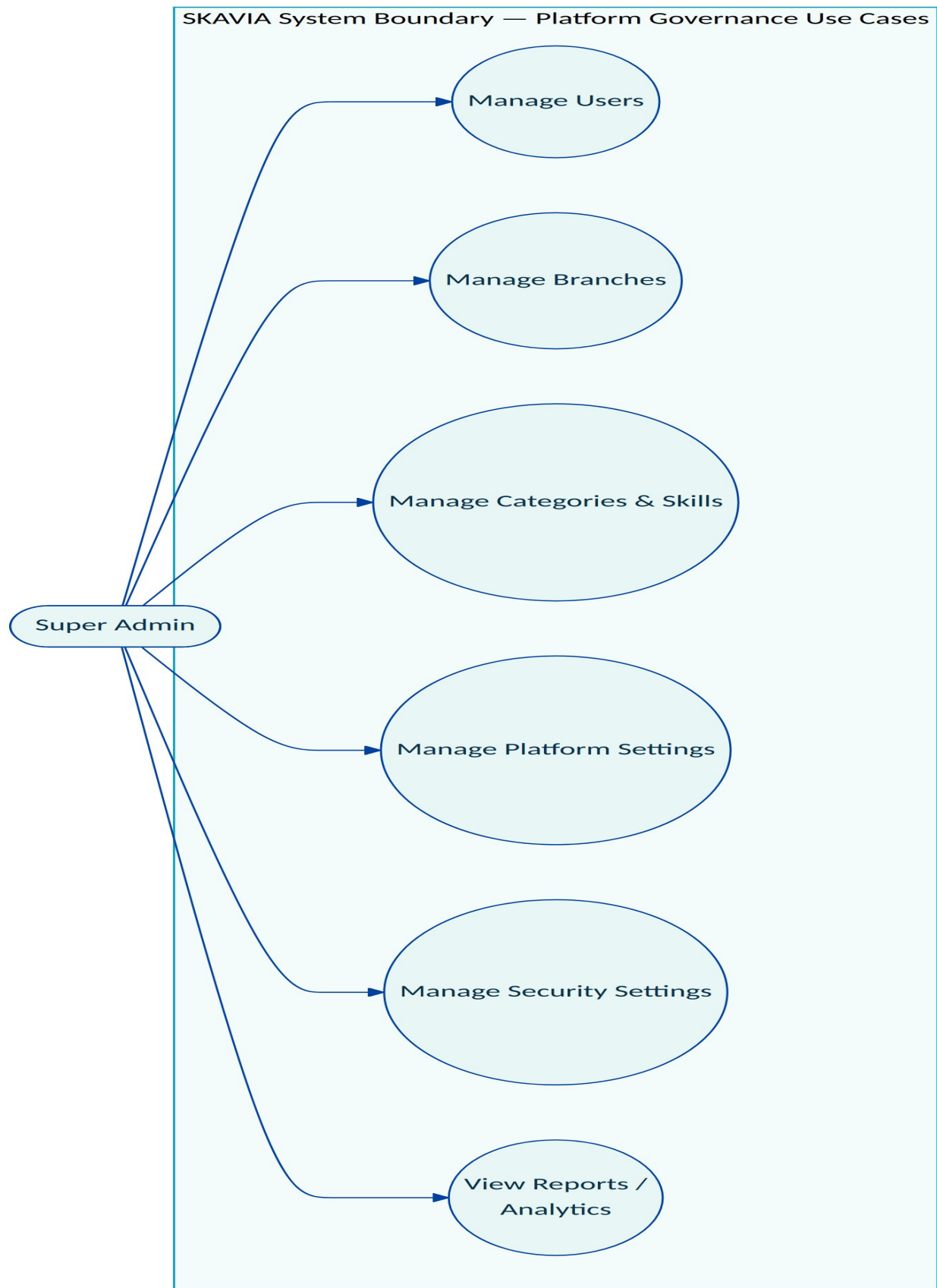


Requesting a Complete Business Solution includes requirement submission, which in turn includes analysis, matching, team creation, Team Leader assignment, and project management in sequence.

4.2.3 Use Case Diagram — Administration

The Administration use case set spans three actors with many shared responsibilities. To keep the diagram readable and avoid excessive line crossings, it is presented as two logically split diagrams: platform governance (Super Admin) and operational administration (Company Admin and Branch Manager), consistent with the role model in Section 2.3.

Figure 4.4a — Administration Use Case Diagram, Part A: Platform Governance



The Super Admin governs platform-wide configuration: users, branches, category/skill taxonomy, platform and security settings, and global reports.

Figure 4.4b — Administration Use Case Diagram, Part B: Operational Administration



Handling a complaint extends into handling a dispute when escalation is required; verifying a worker includes managing the worker record. The Branch Manager's scope is limited to their assigned branch, while the Company Admin's scope spans the platform.

4.3 Use Case Descriptions

The following structured descriptions detail the most important use cases referenced in the diagrams above and throughout the functional requirements.

UC-01 — Register / Login

Field	Detail
Primary Actor(s)	Individual Customer / Worker / Business Client / Organization
Supporting Actor(s)	Authentication Service
Goal	Allow a person to create an account or access an existing account.
Preconditions	The person has a valid email address or phone number.
Trigger	The person selects Register or Login on the application interface.

Main Flow:

1. The person selects Register and enters name, contact information, password, and primary role.
2. The system validates the input and creates a new User record with status Pending Verification.
3. The system sends a verification code to the provided email or phone.
4. The person enters the verification code.
5. The system verifies the code and updates account status to Verified.
6. The person subsequently enters login credentials.
7. The system authenticates the credentials and issues a session/access token.
8. The system redirects the person to their role-based dashboard.

Alternative Flow: If the person already has an account, they proceed directly to step 6 (Login).

Exception Flow: If the verification code is invalid or expired, the system displays an error and allows the person to request a new code. If login credentials are invalid, the system displays an authentication error without indicating whether the email or password was incorrect.

Post conditions: The person holds a valid, authenticated session and can access functions permitted by their role and mode.

UC-02 — Submit Requirement

Field	Detail
Primary Actor(s)	Individual Customer / Business Client / Organization
Supporting Actor(s)	Requirement Management Service
Goal	Allow a client to describe a problem or need without necessarily knowing the exact worker required.
Preconditions	The client is authenticated.
Trigger	The client selects "Submit a Requirement" or an equivalent action.

Main Flow:

1. The client enters a title and a free-text or guided description of the requirement.
2. The client optionally specifies category, location, required date, budget, and attachments.
3. The system assigns a unique requirement identifier and sets status to Submitted.
4. The system forwards the requirement for requirement analysis (see UC-03).
5. The system confirms submission to the client and displays the requirement status.

Alternative Flow: The client may save the requirement as a draft and complete it later, if this is supported by the implementation.

Exception Flow: If mandatory fields are missing, the system displays a validation error and does not submit the requirement.

Post conditions: A new Requirement record exists with status Submitted, ready for analysis.

UC-03 — Requirement Analysis

Field	Detail
Primary Actor(s)	System (Requirement Engine)
Supporting Actor(s)	AI Assistance Layer, Individual Customer / Business Client / Organization
Goal	Convert a natural-language requirement into a structured requirement usable by the Matching Engine.
Preconditions	A requirement has been submitted (UC-02).
Trigger	A new or edited requirement enters the analysis queue.

Main Flow:

1. The Requirement Engine reads the requirement description.
2. The system identifies the relevant category and candidate required skill(s).
3. The system determines whether the requirement is simple (one worker) or complex (multiple skills/team).
4. The system extracts available structured details such as location, date, and budget where present.
5. The system produces a structured requirement record and forwards it to the Matching Engine (UC-04).

Alternative Flow: If essential information is missing, the system prompts the client for clarification before proceeding.

Exception Flow: If the AI-assisted analysis service is unavailable, the system falls back to a manual, form-based structuring method. The system does not fabricate skills, dates, or budget values that were not provided.

Post conditions: A structured requirement exists, linked to identified skill(s) and ready for matching.

UC-04 — Worker Matching

Field	Detail
Primary Actor(s)	System (Matching Engine)
Supporting Actor(s)	SKAVIA Operations / Admin
Goal	Identify and rank suitable candidate workers for a structured requirement.
Preconditions	A structured requirement exists (UC-03).
Trigger	The Requirement Engine forwards a structured requirement to the Matching Engine.

Main Flow:

1. The system retrieves candidate workers whose skills match the requirement's required skill(s).
2. The system calculates a weighted match score for each candidate using skill match, location, availability, experience, rating, verification status, and expected rate.
3. The system excludes candidates who fail mandatory constraints (e.g., unavailable during the requested period).
4. The system produces a ranked candidate list per required skill.

5. For a simple requirement, the ranked list is presented directly to the client. For a complex requirement, the ranked list is presented to SKAVIA operations for team formation (UC-05).

Alternative Flow: If no candidate meets the minimum score threshold, the system notifies the client/operations that no suitable worker was found and may request a widened search radius or relaxed constraints.

Exception Flow: None beyond the alternative flow described above.

Post conditions: A ranked list of Match Result records exists, linked to the requirement.

UC-05 — Request Individual Service

Field	Detail
Primary Actor(s)	Individual Customer
Supporting Actor(s)	Worker, Notification Service
Goal	Allow a customer to book a specific matched worker for a simple requirement.
Preconditions	A ranked worker list exists for the customer's requirement (UC-04).
Trigger	The customer selects a worker from the ranked list.

Main Flow:

1. The customer reviews the worker's profile, ratings, and reviews.
2. The customer selects the worker and requests a booking.
3. The system creates a Service Request record with status Requested and notifies the worker.
4. The worker reviews the request and accepts or rejects it (UC extension).
5. If accepted, the system updates the Service Request status to Accepted and notifies the customer.

Alternative Flow: If the worker rejects the request, the system returns the customer to the ranked candidate list to select another worker.

Exception Flow: If the worker does not respond within the platform's defined response window, the system may automatically expire the request and suggest alternative candidates.

Post conditions: A Service Request exists with a determined status (Accepted, Rejected, or Expired).

UC-06 — Create Team

Field	Detail
Primary Actor(s)	SKAVIA Operations / Admin
Supporting Actor(s)	Matching Engine, Business Client
Goal	Form a team of workers with complementary skills to fulfill a complex requirement.
Preconditions	Ranked candidate lists exist for each required skill of a complex requirement (UC-04).
Trigger	SKAVIA operations confirms the set of workers to be engaged for the requirement.

Main Flow:

1. The administrator reviews ranked candidates for each required skill.
2. The administrator selects one or more workers per required skill.
3. The system creates a Team record and associated Team Member records.
4. The system notifies selected workers of their team assignment.
5. The system proceeds to Team Leader assignment (UC-07).

Alternative Flow: The administrator may manually add or substitute a worker outside the automatically ranked list if operationally justified.

Exception Flow: If an insufficient number of workers is available for a required skill, the system flags the requirement for manual sourcing or client notification.

Post conditions: A Team record exists in status Forming, linked to the requirement.

UC-07 — Assign Team Leader

Field	Detail
Primary Actor(s)	SKAVIA Operations / Admin
Supporting Actor(s)	Team Leader, Notification Service
Goal	Designate one team member as the coordination point for a project.
Preconditions	A Team has been created (UC-06).
Trigger	The administrator selects a Team Leader from the confirmed team members.

Main Flow:

1. The administrator reviews team member profiles, experience, and past leadership performance where available.
2. The administrator assigns the Team Leader role to a selected team member.
3. The system updates the Team record and notifies the assigned Team Leader.
4. The system grants the Team Leader access to project coordination functions (UC-08, UC-09).

Alternative Flow: The client may request a specific worker as Team Leader where supported by the requirement.

Exception Flow: If the assigned Team Leader is unavailable or declines, the administrator selects an alternative team member.

Post conditions: The Team record identifies exactly one active Team Leader.

UC-08 — Create / Manage Project

Field	Detail
Primary Actor(s)	SKAVIA Operations / Admin, Team Leader
Supporting Actor(s)	Business Client, Organization
Goal	Establish and track a project for complex requirement execution.
Preconditions	A Team and Team Leader have been assigned (UC-06, UC-07).
Trigger	The requirement is confirmed for team-based execution.

Main Flow:

1. The system creates a Project record referencing the requirement, client, and assigned team.
2. The Project status is set to Planning.
3. The Team Leader and/or administrator sets or confirms the project deadline.
4. The Project status progresses through Team Formed and In Progress as work begins.
5. The client and Team Leader monitor and update project progress until the client confirms completion, setting status to Completed.

Alternative Flow: The client may request project scope adjustments, subject to administrator/Team Leader review.

Exception Flow: If the client and Team Leader cannot agree on completion, the case may be escalated through the complaint/dispute process (UC-14).

Post conditions: The project reaches a final status of Completed, or remains active with an up-to-date status.

UC-09 — Assign Tasks

Field	Detail
Primary Actor(s)	Team Leader
Supporting Actor(s)	Worker
Goal	Break down project work into trackable units assigned to individual team members.
Preconditions	A Project exists with an assigned Team Leader (UC-08).
Trigger	The Team Leader begins organizing project work.

Main Flow:

1. The Team Leader creates a task with a title, description, assignee, and deadline.
2. The system notifies the assigned worker.
3. The worker updates task progress and marks the task as complete when finished.
4. The Team Leader reviews completed tasks and monitors overall project progress.

Alternative Flow: The Team Leader may reassign a task to a different team member if required.

Exception Flow: If a task deadline passes without completion, the system notifies the Team Leader.

Post conditions: Tasks exist with accurate, current status reflecting project progress.

UC-10 — Complete Service / Project

Field	Detail
Primary Actor(s)	Individual Customer / Business Client, Worker / Team Leader
Supporting Actor(s)	Payment Service, Notification Service
Goal	Formally conclude a service request or project and trigger downstream payment and review steps.
Preconditions	All required work has been performed and submitted.
Trigger	The worker or Team Leader marks work as complete.

Main Flow:

1. The worker/Team Leader marks the service/project as complete.
2. The system notifies the client that confirmation is requested.
3. The client reviews the delivered work and confirms completion.
4. The system updates status to Completed and triggers the payment process (UC-11).

Alternative Flow: The client may request a revision before confirming completion, returning the service/project to In Progress.

Exception Flow: If the client does not respond within a defined period, the platform's completion policy (to be finalized during implementation) determines the next step.

Post conditions: The service/project status is Completed, and payment processing has been triggered.

UC-11 — Make Payment

Field	Detail
Primary Actor(s)	Individual Customer / Business Client
Supporting Actor(s)	Payment Service, Worker
Goal	Process payment for a completed service or project, separating worker payment from the SKAVIA fee.
Preconditions	A service/project has been confirmed complete (UC-10).
Trigger	Completion confirmation triggers the payment workflow.

Main Flow:

1. The system calculates the total payment as worker payment plus SKAVIA service fee.
2. The client completes payment through a supported payment method.
3. The system records the Payment and Transaction with status Completed.
4. The system creates a Worker Earning record for the worker's portion.
5. The system generates an invoice/receipt and notifies both parties.

Alternative Flow: None beyond standard payment method selection.

Exception Flow: If payment fails, the system records status Failed and notifies the client to retry.

Post conditions: A completed Payment record exists, and the worker's earnings have been updated.

UC-12 — Submit Review

Field	Detail
Primary Actor(s)	Individual Customer / Business Client
Supporting Actor(s)	Worker
Goal	Capture client feedback and update worker reputation after completion.
Preconditions	Payment has been completed (UC-11), or completion has been confirmed.
Trigger	The client is prompted to review the worker/team after completion.

Main Flow:

1. The client submits a numeric rating and an optional written review.
2. The system stores the Review record linked to the worker and the service/project.
3. The system recalculates the worker's aggregated reputation score.
4. The system notifies the worker of the new review.

Alternative Flow: The client may decline to leave a written comment while still submitting a rating.

Exception Flow: The system rejects review submission attempts made before confirmed completion.

Post conditions: The worker's profile reflects the updated rating, review count, and reputation.

UC-13 — Worker Verification

Field	Detail
Primary Actor(s)	Worker
Supporting Actor(s)	Company Admin / Branch Manager
Goal	Establish trust by confirming a worker's identity, skill, or certificate claims.
Preconditions	The worker has an active profile.
Trigger	The worker submits verification information.

Main Flow:

1. The worker submits identity, certificate, or experience information relevant to their skill category.
2. The system creates a Verification record with status Pending and notifies administration.
3. An authorized administrator reviews the submitted information.
4. The administrator approves, rejects, or requests additional information.
5. The system updates the worker's verification status and notifies the worker of the outcome.

Alternative Flow: If additional information is requested, the worker resubmits, restarting from step 3.

Exception Flow: If verification is rejected, the worker profile retains an unverified status for that verification type.

Post conditions: The Worker Profile displays an accurate, current verification status.

UC-14 — Complaint / Dispute Management

Field	Detail
Primary Actor(s)	Individual Customer / Worker / Business Client
Supporting Actor(s)	Company Admin / Branch Manager
Goal	Provide a structured channel for reporting and resolving issues related to services or projects.
Preconditions	A service request or project exists that the complainant wishes to report an issue about.
Trigger	A user submits a complaint.

Main Flow:

1. The user submits a complaint with category, description, and optional attachments, referencing the relevant service/project.
2. The system creates a Complaint record with status Open and notifies administration.
3. An authorized administrator reviews the complaint and updates status to Under Review.
4. The administrator resolves the complaint directly, or escalates it to a formal Dispute for investigation.
5. The administrator records a resolution and updates status to Resolved, then Closed.
6. The system notifies the complainant of the outcome at each significant status change.

Alternative Flow: A resolved complaint may be reopened if the complainant disputes the resolution, subject to platform policy.

Exception Flow: If a dispute cannot be resolved through the standard workflow, it remains open for extended investigation by SKAVIA management.

Post conditions: The complaint/dispute reaches a recorded, notified resolution or remains actively tracked.

UC-15 — Administrative Management (User/Worker Management)

Field	Detail
Primary Actor(s)	Super Admin / Company Admin / Branch Manager
Supporting Actor(s)	None
Goal	Allow authorized administration roles to manage platform users, workers, and operational data within their permission scope.
Preconditions	The administrator is authenticated with the appropriate role.
Trigger	The administrator accesses the relevant administration dashboard module.

Main Flow:

1. The administrator views a list of users, workers, businesses, or organizations within their permission scope.
2. The administrator searches or filters the list as needed.
3. The administrator views details of a selected record.
4. The administrator performs a permitted action, such as activation, deactivation, verification review, or data correction.
5. The system records the action in the audit log where applicable.

Alternative Flow: A Branch Manager's view and actions are scoped to their assigned branch, while a Company Admin's view spans the platform.

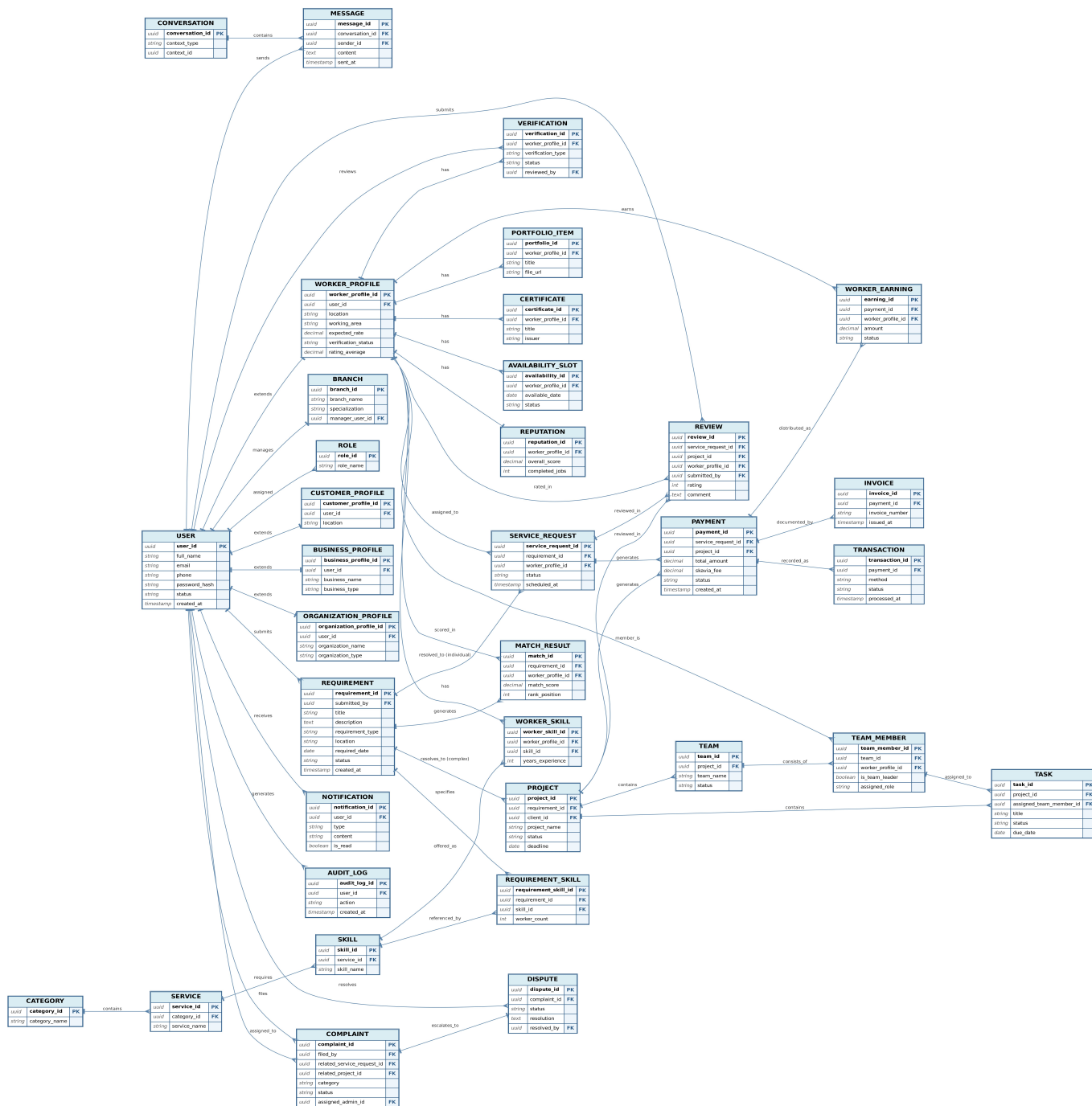
Exception Flow: If the administrator attempts an action outside their permission scope, the system denies the action and displays an authorization error.

Post conditions: The relevant record is updated, and the action is logged where applicable.

4.4 Entity Relationship (ER) Diagram

The following ER diagram reflects the actual entities required to support SKAVIA's functional requirements: identity and profiles, category/service/skill taxonomy, requirement structuring and matching, individual service requests, team/project/task execution, payment and earnings, reviews and reputation, communication, notifications, and complaint/dispute handling. This model is implemented as a relational schema within Supabase's PostgreSQL-compatible database. Every entity shown supports at least one functional requirement in Section 3.1, and every major relationship in Section 3.1 is represented in this model.

Figure 4.5 — SKAVIA Entity Relationship Diagram



Many-to-many relationships (e.g., Worker ↔ Skill, Requirement ↔ Skill, Team ↔ Worker) are resolved through associative entities: WORKER_SKILL, REQUIREMENT_SKILL, and TEAM_MEMBER respectively.

4.5 Data Dictionary

The data dictionary below documents the primary attributes of the most significant entities shown in the ER diagram. Composite/child entities (e.g., Team and Team Member) are documented together where their attributes are closely related. Generic, PostgreSQL-compatible data types are used, consistent with the confirmed Supabase database platform; exact column-level constraints are to be finalized during implementation.

USER

Attribute	Data Type	Key	Description
user_id	UUID	PK	Unique identifier for the account holder.
full_name	VARCHAR		Full name of the person.
email	VARCHAR		Email address; used for login and verification.
phone	VARCHAR		Phone number; alternate login/verification channel.
password_hash	VARCHAR		Cryptographically hashed password.
status	VARCHAR		Account status (Pending Verification, Active, Deactivated).
created_at	TIMESTAMP		Record creation timestamp.

WORKER_PROFILE

Attribute	Data Type	Key	Description
worker_profile_id	UUID	PK	Unique identifier for the worker profile.
user_id	UUID	FK → USER	Owning user account.
location	VARCHAR		Primary location of the worker.
working_area	VARCHAR		Geographic area the worker serves.
expected_rate	DECIMAL		Worker's expected rate for services.
verification_status	VARCHAR		Overall verification status.
rating_average	DECIMAL		Aggregated average rating.

REQUIREMENT

Attribute	Data Type	Key	Description
requirement_id	UUID	PK	Unique identifier for the requirement.
submitted_by	UUID	FK → USER	User who submitted the requirement.
title	VARCHAR		Short title of the requirement.
description	TEXT		Free-text description of the need.
requirement_type	VARCHAR		Simple or Complex classification.
location	VARCHAR		Location relevant to the requirement.
required_date	DATE		Date or period the service/project is needed.
status	VARCHAR		Current status of the requirement.
created_at	TIMESTAMP		Record creation timestamp.

MATCH_RESULT

Attribute	Data Type	Key	Description
match_id	UUID	PK	Unique identifier for a match record.
requirement_id	UUID	FK → REQUIREMENT	Requirement being matched.

Attribute	Data Type	Key	Description
worker_profile_id	UUID	FK → WORKER_PROFILE	Candidate worker.
match_score	DECIMAL		Calculated weighted match score.
rank_position	INTEGER		Rank of this candidate for the requirement.

SERVICE_REQUEST

Attribute	Data Type	Key	Description
service_request_id	UUID	PK	Unique identifier for an individual service booking.
requirement_id	UUID	FK → REQUIREMENT	Originating requirement.
worker_profile_id	UUID	FK → WORKER_PROFILE	Assigned worker.
status	VARCHAR		Requested, Accepted, Rejected, In Progress, Completed, Canceled.
scheduled_at	TIMESTAMP		Scheduled date/time for the service.

TEAM / TEAM_MEMBER

Attribute	Data Type	Key	Description
team_id	UUID	PK	Unique identifier for a team.
project_id	UUID	FK → PROJECT	Project the team is assigned to.
status	VARCHAR		Forming, Active, Completed.
team_member_id	UUID	PK (child)	Unique identifier for a membership record.
worker_profile_id	UUID	FK → WORKER_PROFILE	Member worker.
is_team_leader	BOOLEAN		Indicates whether this member is the Team Leader.

PROJECT / TASK

Attribute	Data Type	Key	Description
project_id	UUID	PK	Unique identifier for a project.
requirement_id	UUID	FK → REQUIREMENT	Originating requirement.
client_id	UUID	FK → USER	Client who owns the project.
status	VARCHAR		Requested, Planning, Team Formed, In Progress, Review, Completed.
deadline	DATE		Target completion date.
task_id	UUID	PK (child)	Unique identifier for a task.
assigned_team_member_id	UUID	FK → TEAM_MEMBER	Worker assigned to the task.
status	VARCHAR		Assigned, In Progress, Submitted, Completed.

PAYMENT / WORKER_EARNING

Attribute	Data Type	Key	Description
payment_id	UUID	PK	Unique identifier for a payment.
service_request_id / project_id	UUID	FK	Related service/project.
total_amount	DECIMAL		Total amount charged to the client.
skavia_fee	DECIMAL		SKAVIA's service/management fee portion.
status	VARCHAR		Pending, Completed, Failed, Refunded.
earning_id	UUID	PK (child)	Unique identifier for a worker earning record.
amount	DECIMAL		Worker's portion of the payment.

REVIEW / REPUTATION

Attribute	Data Type	Key	Description
review_id	UUID	PK	Unique identifier for a review.
worker_profile_id	UUID	FK → WORKER_PROFILE	Worker being reviewed.
rating	INTEGER		Numeric rating value.
comment	TEXT		Written review text.
reputation_id	UUID	PK (child)	Unique identifier for the reputation record.
overall_score	DECIMAL		Aggregated reputation score.
completed_jobs	INTEGER		Number of completed services/projects.

COMPLAINT / DISPUTE

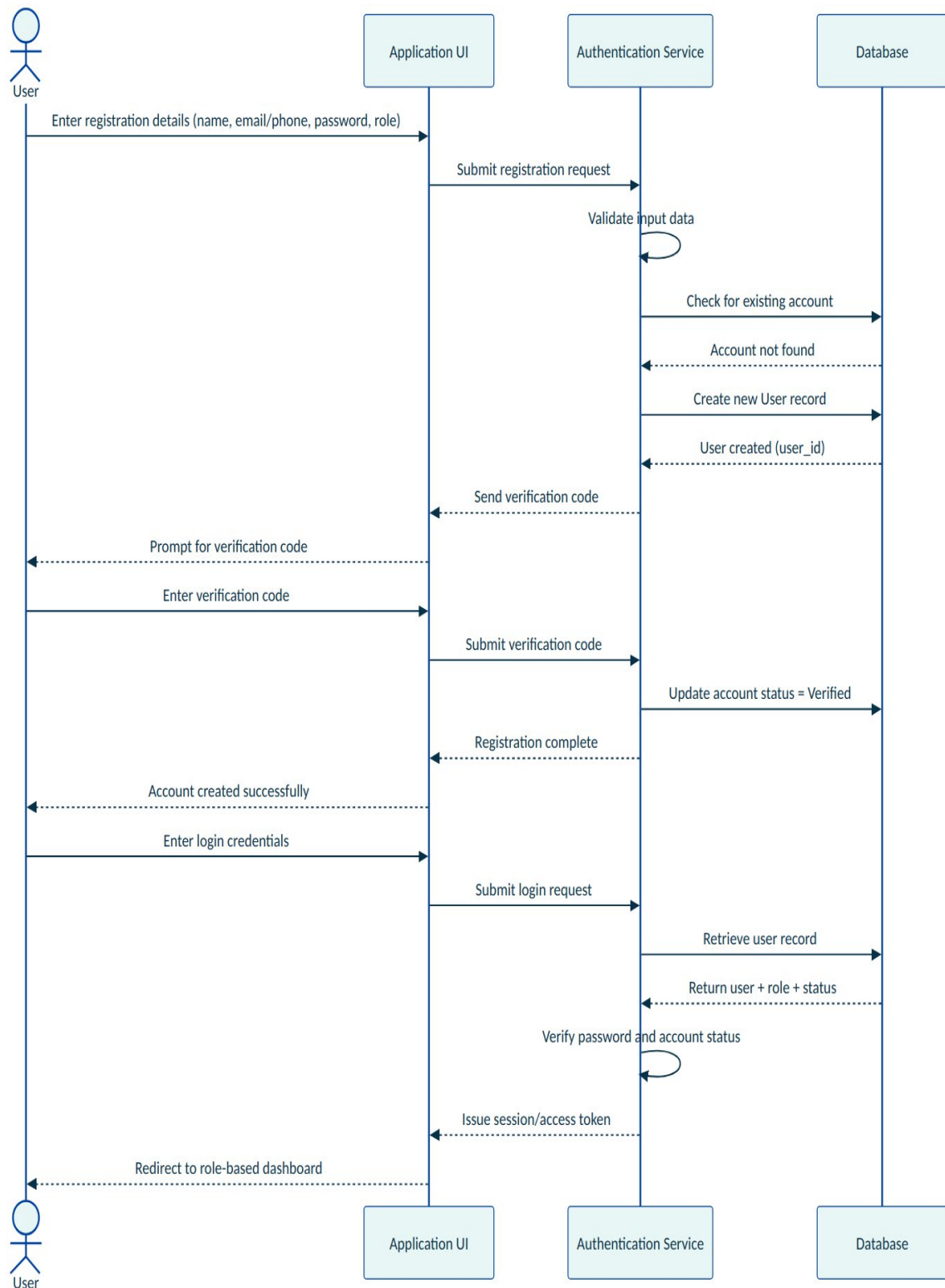
Attribute	Data Type	Key	Description
complaint_id	UUID	PK	Unique identifier for a complaint.
filed_by	UUID	FK → USER	User who filed the complaint.
category	VARCHAR		Complaint category.
status	VARCHAR		Open, Under Review, Assigned, Resolved, Closed.
dispute_id	UUID	PK (child)	Unique identifier for an escalated dispute.
resolution	TEXT		Recorded resolution decision.

4.6 Sequence Diagrams

The following sequence diagrams illustrate the most important SKAVIA workflows. Lifelines represent the actor(s) involved and the system components each workflow passes through: the Application UI, Backend Service, Requirement Engine, Matching Engine, Database, Payment Service, Notification Service, and Administration, as applicable. Each diagram is consistent with the functional requirements referenced in its caption and with the Requirement Traceability Matrix in Section 4.8.

4.6.1 Sequence 1 — User Registration / Login

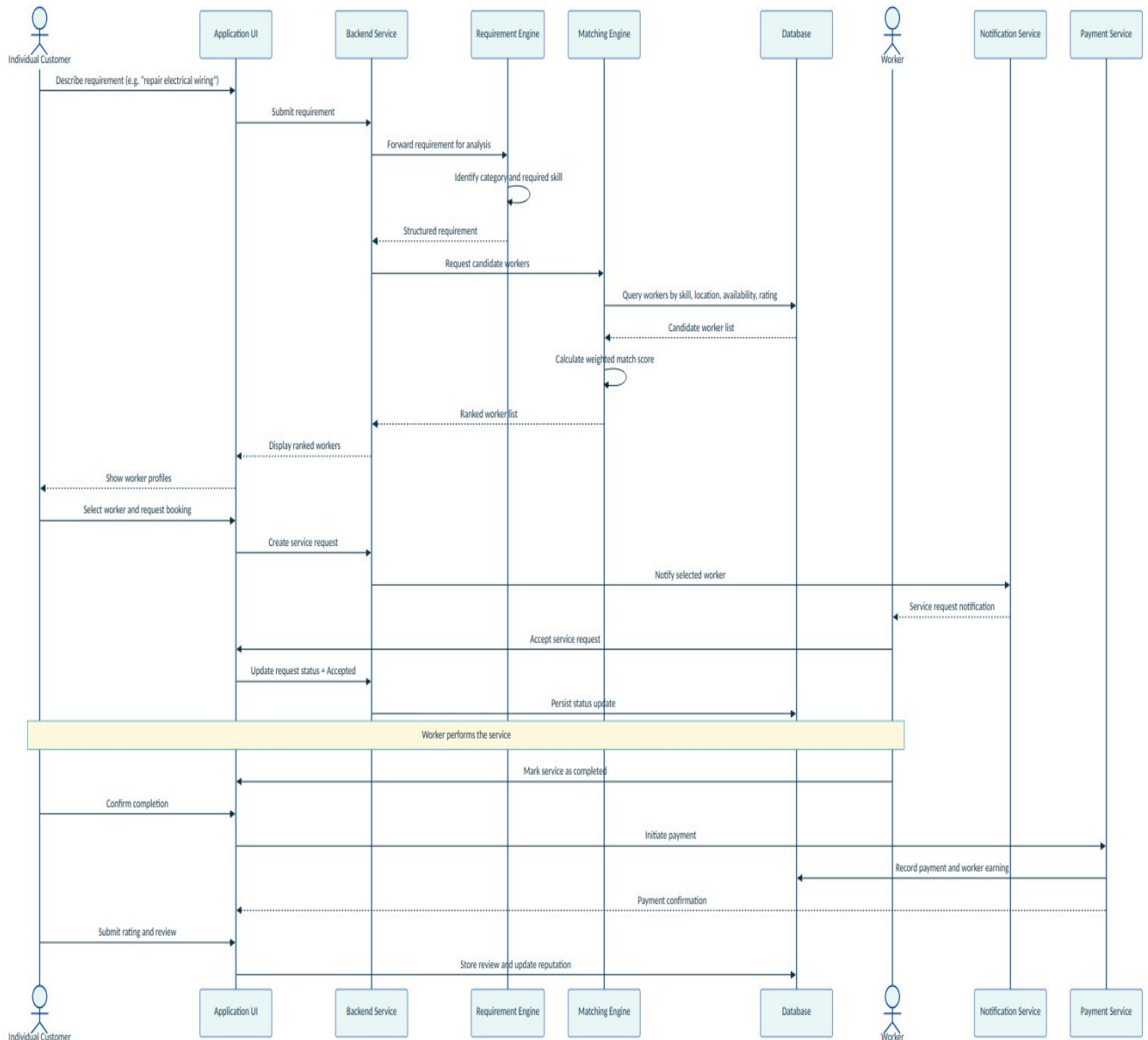
Figure 4.6 — Registration and Login Sequence



Implements FR-A01 through FR-A03: account creation, verification, and authentication.

4.6.2 Sequence 2 — Individual Service Request

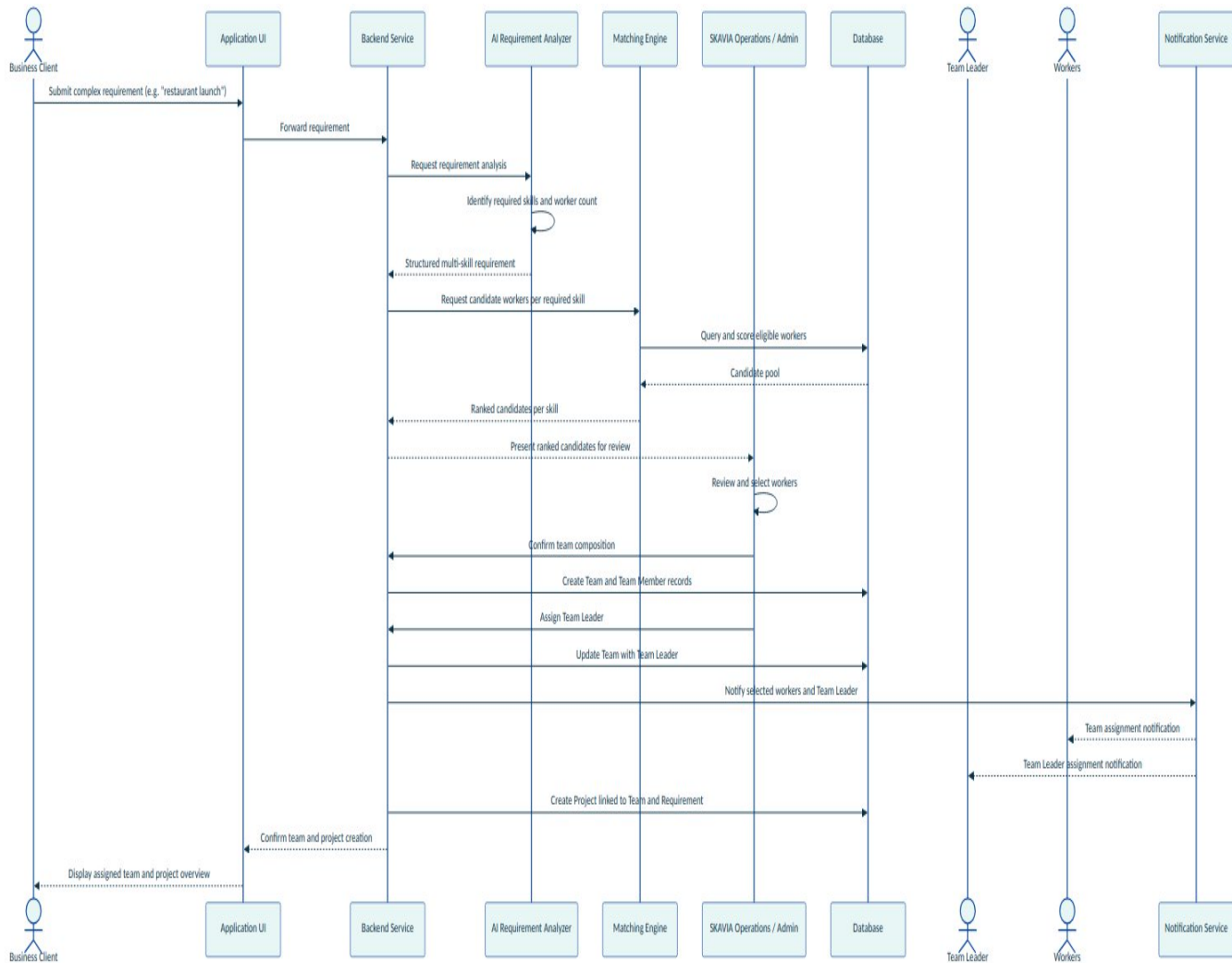
Figure 4.7 — Individual Service Request Sequence



Implements FR-E01–E05, FR-F01–F04, FR-G01–G03, and FR-H01–H05: requirement submission through matching, booking, execution, and initial payment/review trigger.

4.6.3 Sequence 3 — Complex Requirement to Team Formation

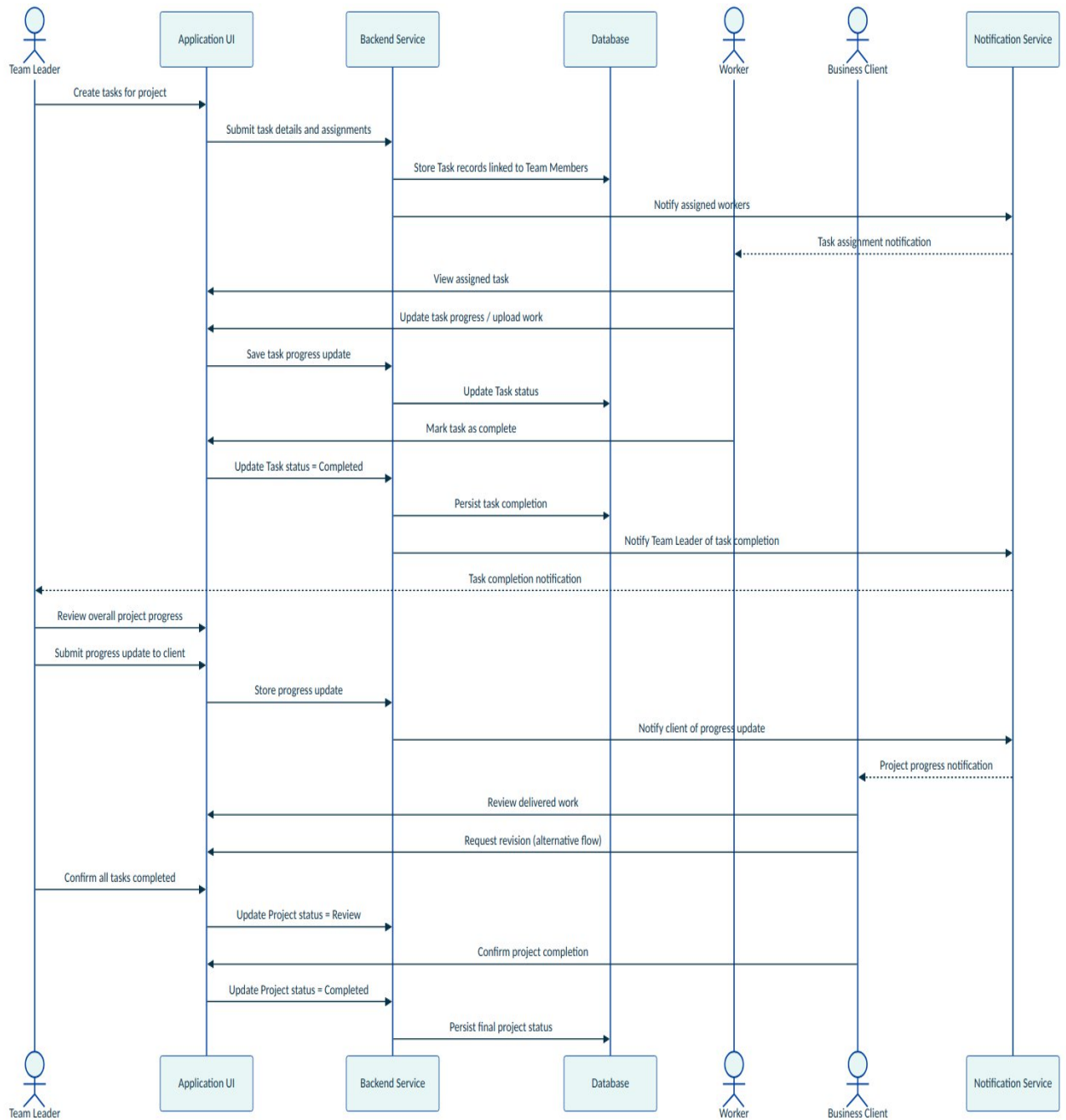
Figure 4.8 — Complex Requirement, Matching, and Team Formation Sequence



Implements FR-F01–F06, FR-G01–G05, FR-I01–I06, and FR-J01: requirement analysis identifying multiple required skills, ranked candidate matching per skill, operations-confirmed team creation, and Team Leader assignment.

4.6.4 Sequence 4 — Project Execution, Task Assignment, and Completion

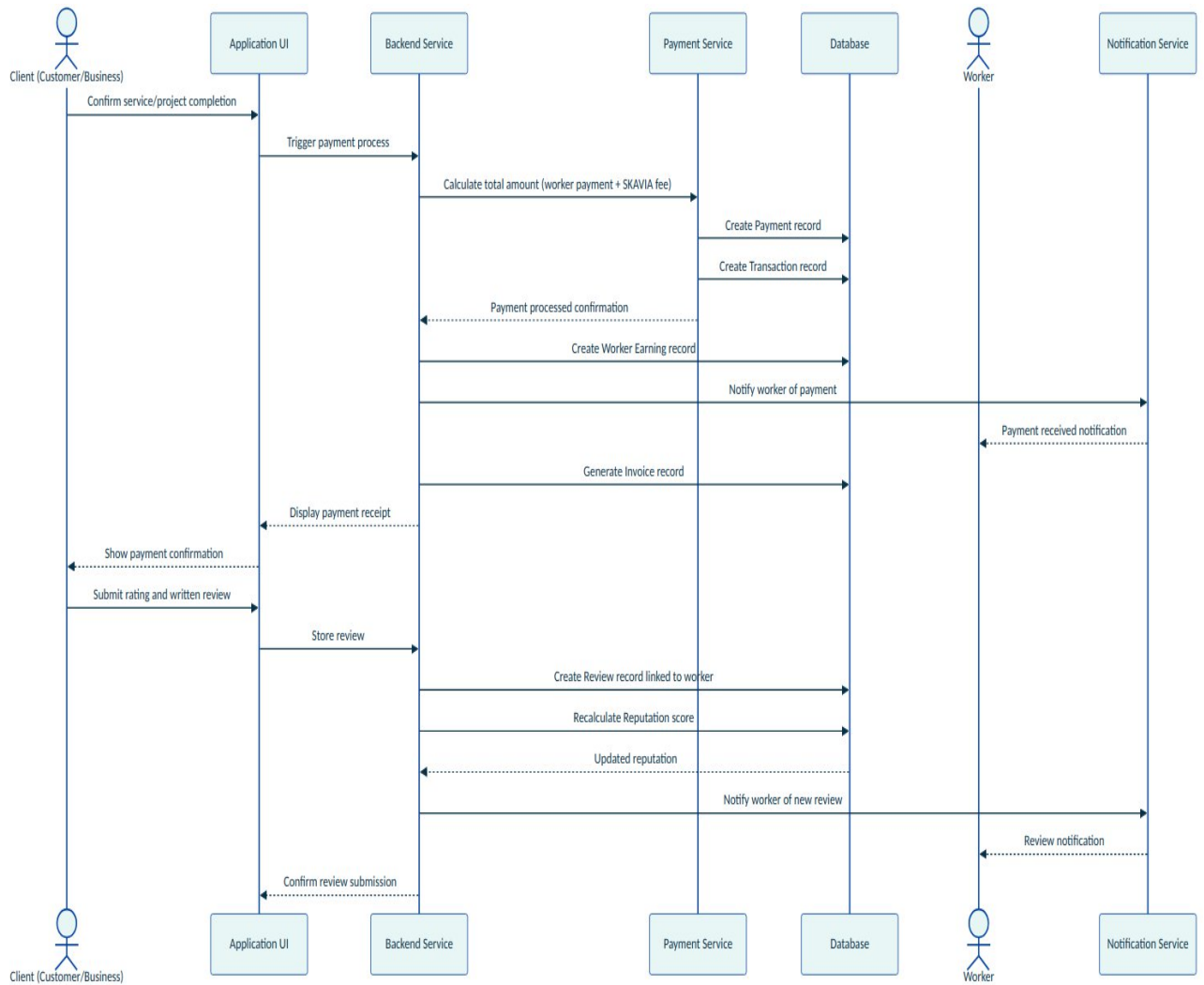
Figure 4.9 — Project Execution Sequence



Implements FR-J02–J07, FR-K01–K06, and FR-L01–L05: task assignment, progress updates, and project completion confirmation.

4.6.5 Sequence 5 — Payment, Completion, Worker Earnings, and Review

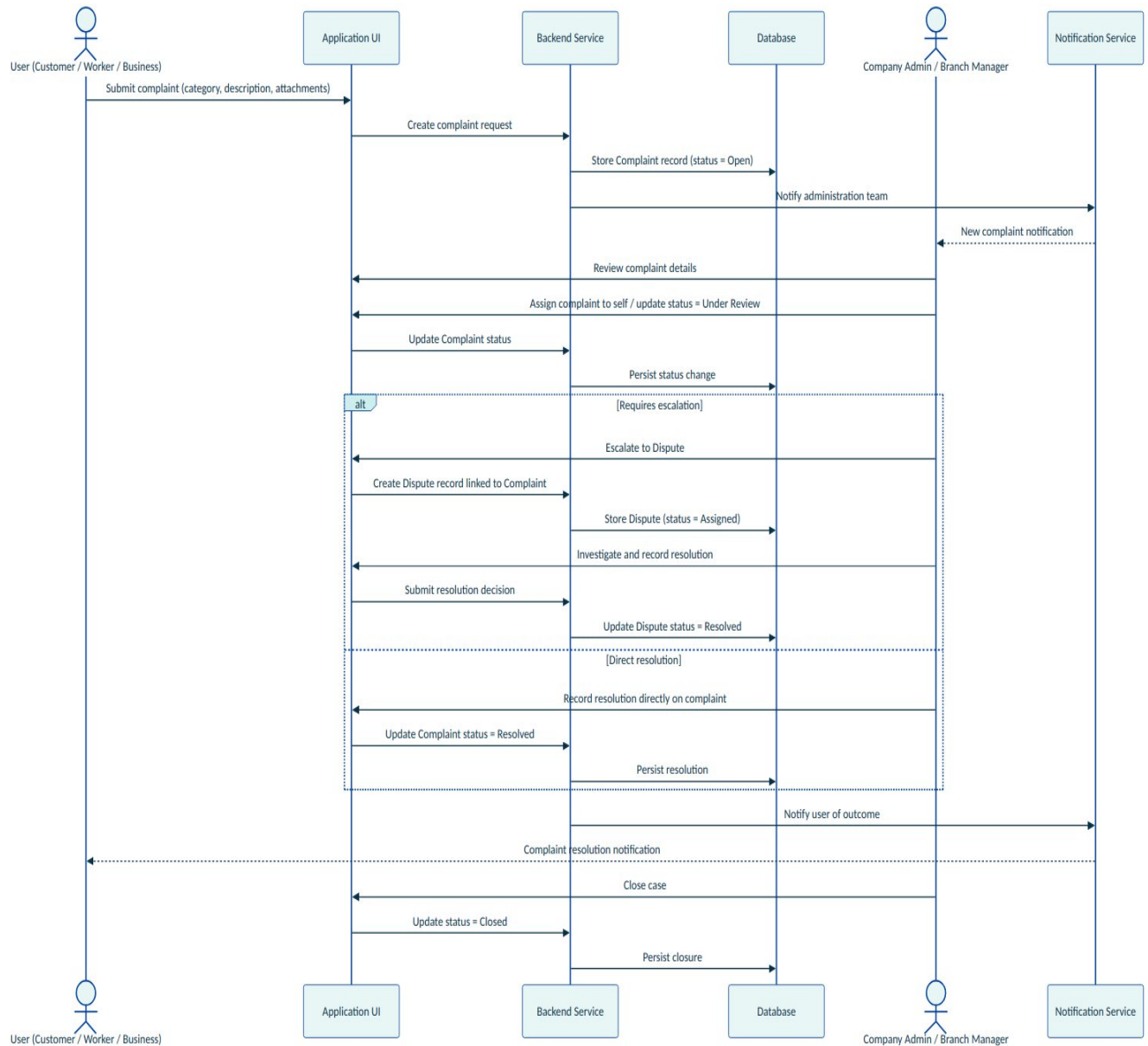
Figure 4.10 — Payment and Review Sequence



Implements FR-O01–O05, FR-P01–P04, and FR-Q01–Q04: fee-separated payment calculation, worker earning recording, invoicing, and review-triggered reputation update.

4.6.6 Sequence 6 — Complaint / Dispute Handling

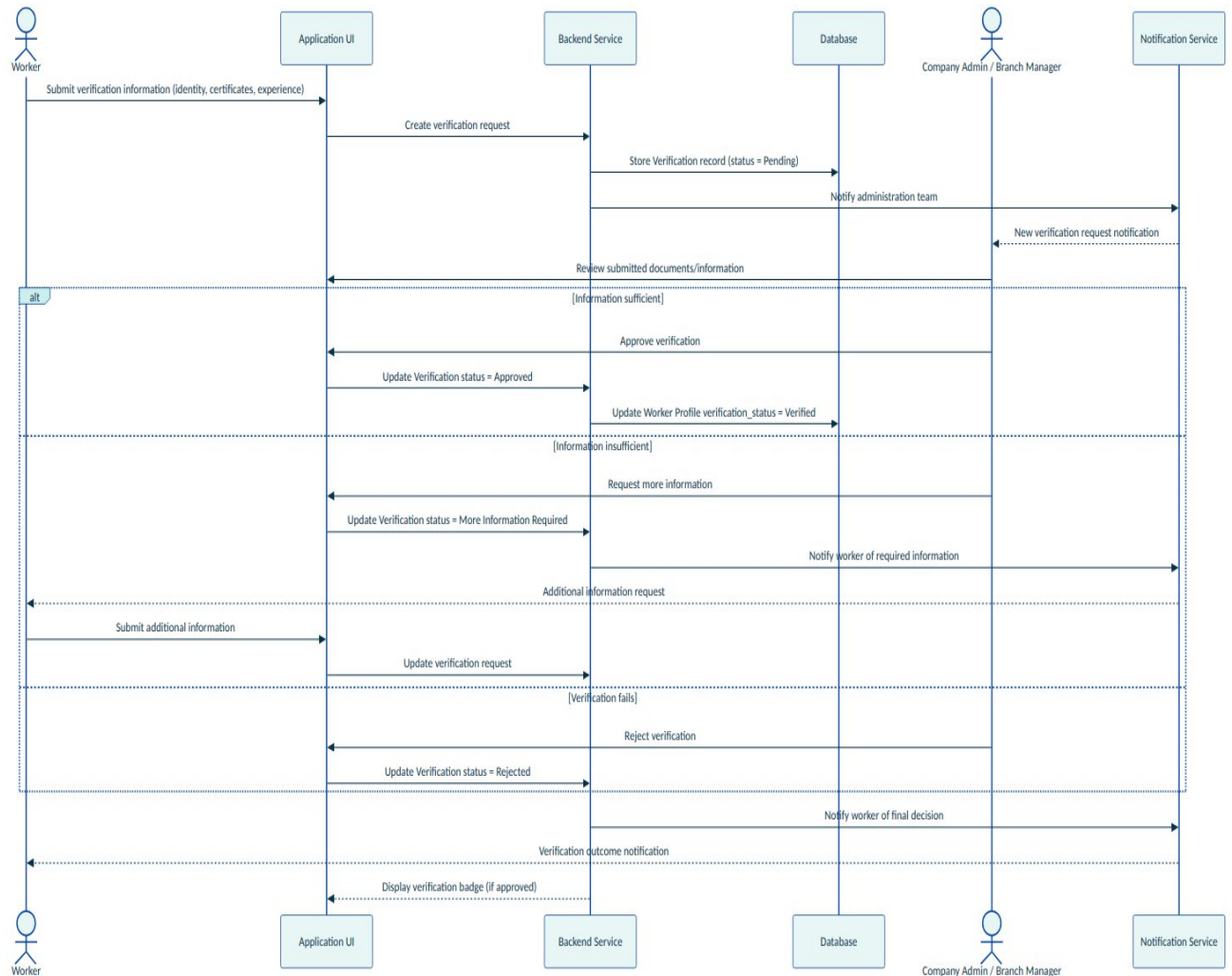
Figure 4.11 — Complaint and Dispute Sequence



Implements FR-S01–FR-S05: complaint submission, administrative review, optional escalation to a formal dispute, resolution, and closure.

4.6.7 Sequence 7 — Worker Verification

Figure 4.12 — Worker Verification Sequence



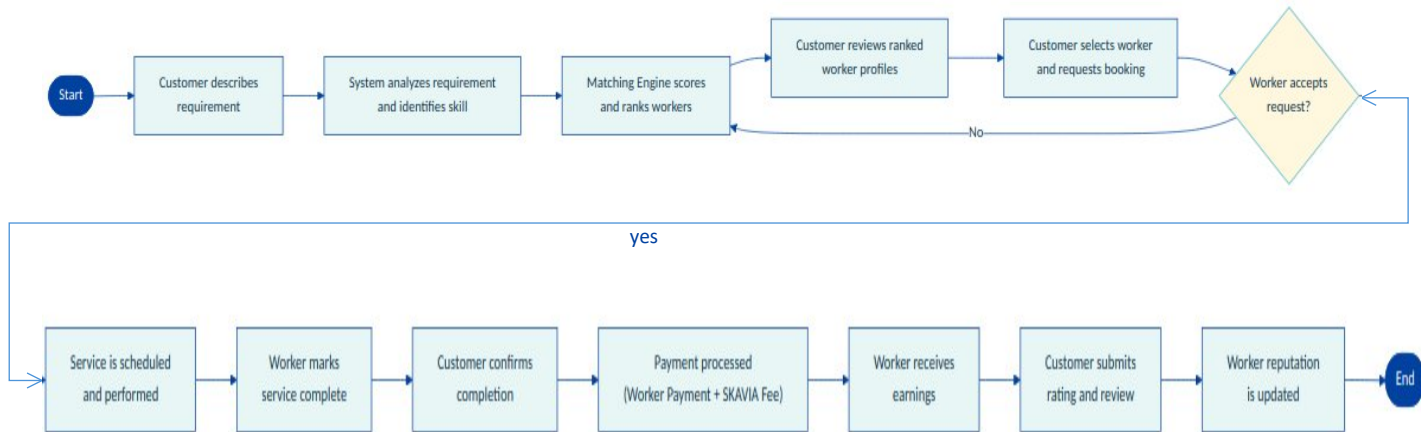
Implements FR-R01–FR-R04: verification submission, administrative review with three possible outcomes, and profile status update.

4.7 Activity / System Flow Diagrams

The following activity diagrams present five important end-to-end system flows in a process-oriented form, complementing the actor/message-oriented sequence diagrams above. Team formation and project management, which were previously combined, are now presented as two distinct, more detailed flows for clarity, alongside a high-level Complete Business Solution Flow that shows how they connect.

4.7.1 Individual Service Flow

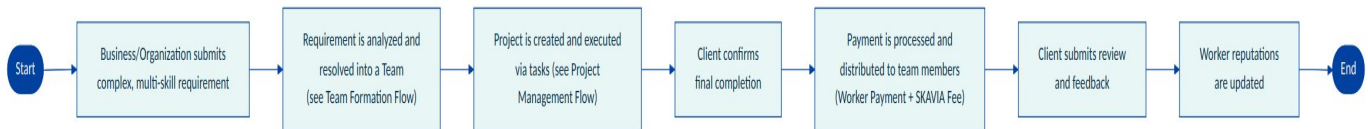
Figure 4.13 — Individual Service Flow



Corresponds to Use Case UC-05 and Sequence Diagram 2.

4.7.2 Complete Business Solution Flow

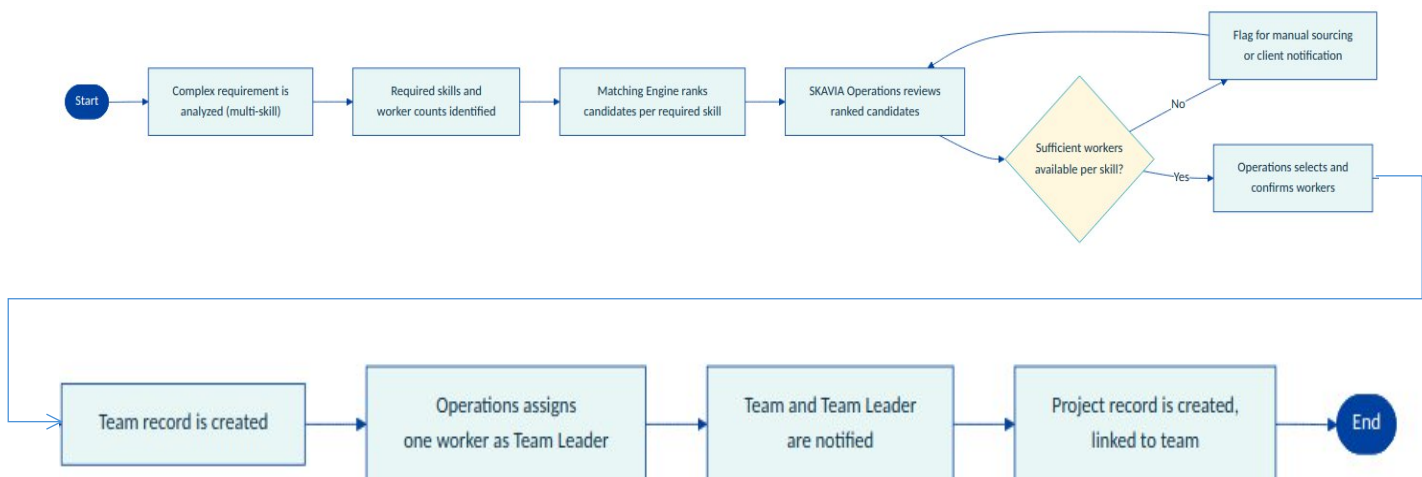
Figure 4.14 — Complete Business Solution Flow



A high-level overview corresponding to Use Cases UC-06 through UC-11; the two shaded steps are detailed further in the Team Formation Flow (Figure 4.15) and Project Management Flow (Figure 4.16).

4.7.3 Team Formation Flow

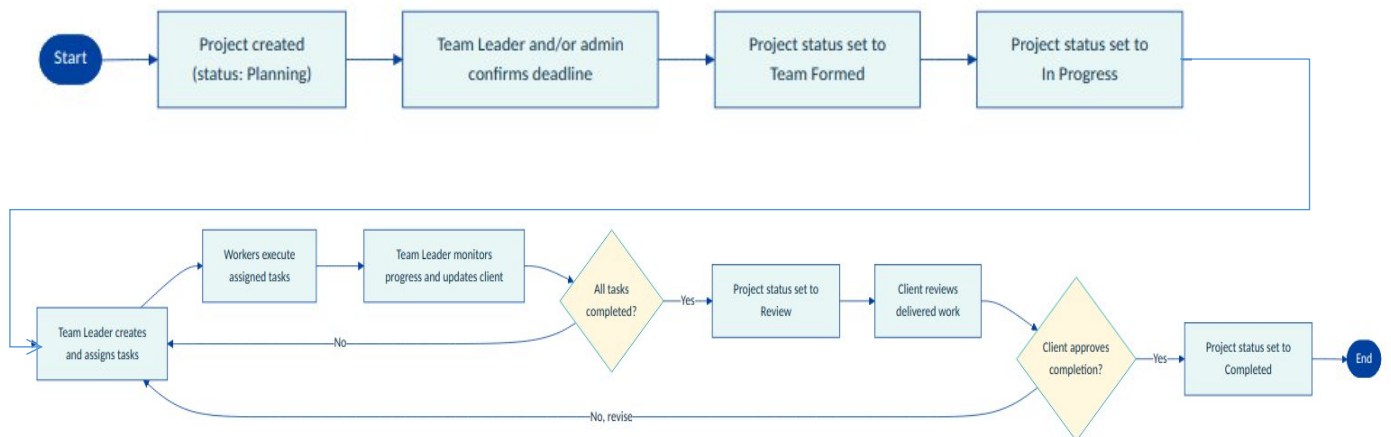
Figure 4.15 — Team Formation Flow



Corresponds to Use Cases UC-04, UC-06, and UC-07, and Sequence Diagram 3.

4.7.4 Project Management Flow

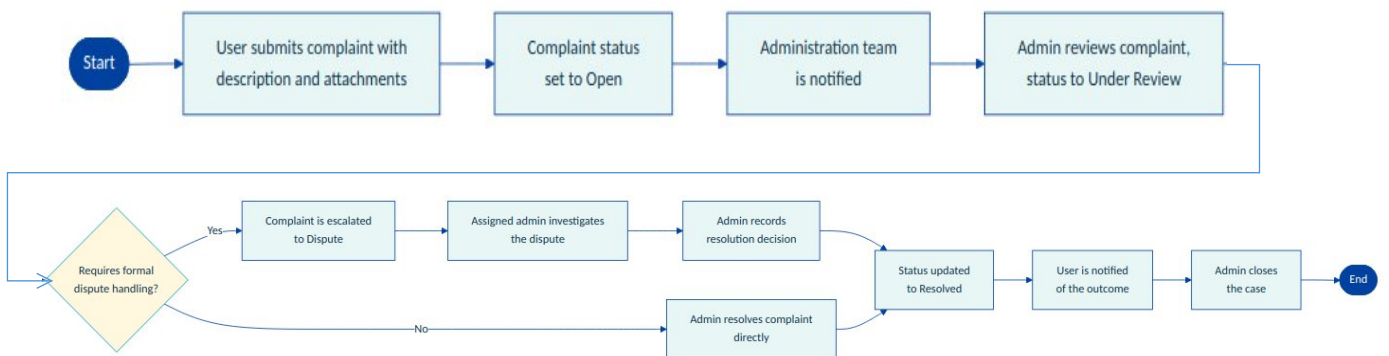
Figure 4.16 — Project Management Flow



Corresponds to Use Cases UC-08 and UC-09, and Sequence Diagram 4.

4.7.5 Complaint / Dispute Flow

Figure 4.17 — Complaint / Dispute Flow



Corresponds to Use Case UC-14 and Sequence Diagram 6.

4.8 Requirement Traceability Matrix

The table below connects representative functional requirement identifiers to their corresponding use case, functional module, database entity, and system diagram(s), demonstrating that every major functional area of this SRS is reflected consistently across the requirements, use case model, database model, and behavioral diagrams.

Requirement ID(s)	Use Case	Module	Database Entity	Diagram(s)
FR-A01	UC-01 Register / Login	Authentication & Account	USER, ROLE	Sequence 1; Use Case Diagram 1
FR-A08	UC-15 Administrative Management	Authentication & Account	USER, ROLE	Architecture Diagram
FR-A09/A10	UC-01 Register / Login	Authentication & Account	USER, WORKER_PROFILE, CUSTOMER_PROFILE	Architecture Diagram
FR-B02	Manage Profile (part of UC-01 flow)	Profile Management	WORKER_PROFILE, WORKER_SKILL, PORTFOLIO_ITEM,	ER Diagram; Use Case Diagram 1

Requirement ID(s)	Use Case	Module	Database Entity	Diagram(s)
			CERTIFICATE	
FR-B08	Manage Profile	Profile Management	AVAILABILITY_SLOT	ER Diagram
FR-C01–C03	Manage Categories & Skills (UC-15)	Category & Service Mgmt	CATEGORY, SERVICE, SKILL	ER Diagram; Use Case Diagram 3a
FR-D01–D05	Search Services / Workers	Search	SKILL, SERVICE, WORKER_PROFILE	Use Case Diagram 1
FR-E01–E05	UC-02 Submit Requirement	Requirement Submission	REQUIREMENT	Sequence 2, 3; Individual Service Flow, Complete Business Solution Flow
FR-F01–F06	UC-03 Requirement Analysis	Requirement Analysis	REQUIREMENT, REQUIREMENT_SKILL	Sequence 2, 3; Architecture Diagram
FR-G01–G05	UC-04 Worker Matching	Worker Matching	MATCH_RESULT, WORKER_SKILL	Sequence 2, 3; ER Diagram; Team Formation Flow
FR-H01–H05	UC-05 Request Individual Service	Individual Services	SERVICE_REQUEST	Sequence 2; Individual Service Flow
FR-I01–I06	UC-06 Create Team	Team Management	TEAM, TEAM_MEMBER	Sequence 3; ER Diagram; Team Formation Flow
FR-J01–J07	UC-07 Assign Team Leader	Team Leader	TEAM_MEMBER (is_team_leader)	Sequence 3, 4; Team Formation Flow, Project Management Flow
FR-K01–K06	UC-08 Create / Manage Project	Project Management	PROJECT	Sequence 4; Project Management Flow
FR-L01–L05	UC-09 Assign Tasks	Task Management	TASK	Sequence 4; Project Management Flow
FR-M01–M05	Communicate (supporting use case)	Communication	CONVERSATION, MESSAGE	ER Diagram; Architecture Diagram
FR-N01–N07	Receive Notifications (supporting use case)	Notifications	NOTIFICATION	All Sequence Diagrams; ER Diagram
FR-O01–O05	UC-11 Make Payment	Payment	PAYMENT, TRANSACTION, INVOICE	Sequence 5; Individual Service Flow, Complete Business Solution Flow
FR-P01–P04	UC-11 Make Payment	Worker Earnings	WORKER_EARNING	Sequence 5
FR-Q01–Q04	UC-12 Submit Review	Reviews & Reputation	REVIEW, REPUTATION	Sequence 5; Individual Service Flow, Complete Business Solution Flow
FR-R01–R04	UC-13 Worker Verification	Verification	VERIFICATION	Sequence 7
FR-S01–S05	UC-14 Complaint / Dispute Management	Complaints & Disputes	COMPLAINT, DISPUTE	Sequence 6; Complaint / Dispute Flow
FR-T01–T07	UC-15 Administrative Management	Administration	USER, AUDIT_LOG, BRANCH	Use Case Diagrams 3a, 3b; Architecture Diagram
FR-U01–U03	View Reports / Analytics (supporting use case)	Reporting & Analytics	(aggregated across entities)	Use Case Diagrams 3a, 3b

4.9 Application Module Structure (Dashboards by Role)

The table below summarizes the primary dashboard modules available to each role, consistent with the role/permission model in Section 2.3 and the functional requirements in Section 3.1. Only modules supported by the source project material are listed; no additional screens are introduced beyond what the specified functionality requires.

Role	Primary Dashboard Modules
Worker	Profile & Skills, Availability Calendar, Portfolio & Certificates, Incoming Requests, My Services/Team Assignments, Tasks, Messages, Earnings, Reviews, Verification Status, Notifications.
Individual Customer	Profile, Submit Requirement, Search Services/Workers, My Bookings, Messages, Payments, Reviews, Notifications.
Business Client	Profile, Submit Requirement / Request Complete Business Solution, My Projects, Team & Progress Overview, Messages, Payments, Reviews, Reports, Notifications.
Organization	Profile, Submit Requirement, My Projects, Reports, Messages, Notifications (functionally aligned with Business Client).
Team Leader	Assigned Project Overview, Team Composition, Task Assignment & Tracking, Progress Reporting, Client Communication, Issue Reporting, Notifications.
Branch Manager	Branch Dashboard, Branch Worker Management, Branch Requirement/Project/Team Monitoring, Verification Queue (branch scope), Complaints (branch scope), Branch Reports.
Company Admin	Operations Dashboard, User/Worker/Business Management, Verification Queue, Payments & Fee Monitoring, Complaints & Disputes, Platform Reports.
Super Admin	Platform Dashboard, Company Admin Management, Branch Management, Category/Service/Skill Management, Platform Settings, Security Settings, Platform-Wide Reports.

5. Project Management

5.1 Development Tools

The tools listed below reflect the categories of tooling required to build the system described in this SRS. Where the source project documents do not specify an exact technology, this is stated explicitly rather than invented.

Tool / Technology Category	Specific Tool	Usage in SKAVIA
Requirements & Design	UML diagramming (use case, ER, sequence, activity)	Modeling the system structure and workflows described in this SRS.
Version Control	Git-based repository hosting	Source code management and collaboration.
Mobile Application Framework	Flutter (Dart)	Building the single cross-platform mobile application interface used by all user roles.
Programming Language	Dart	Primary language for implementing the Flutter mobile application.
Backend-as-a-Service Platform	Supabase (PostgreSQL database, auto-generated APIs, and realtime capabilities)	Persistent storage and data access for users, requirements, matches, teams, projects, payments, and reviews, consistent with the ER model in Section 4.4.

Tool / Technology Category	Specific Tool	Usage in SKAVIA
Authentication Service	Supabase Auth	Handling account registration, login, session/token management, and verification, consistent with functional requirement group A (Section 3.1.A).
File / Image Storage	Firebase Storage	Storing worker portfolio images, certificates, attachments, and other media referenced by the Worker Profile, Requirement, and Complaint modules.
Development IDE	Android Studio and Visual Studio Code	Primary development environments for building and debugging the Flutter mobile application.
Payment Gateway	To be finalized during implementation	Processing client payments and worker payouts.
Notification Delivery	To be finalized during implementation	Delivering in-app and/or push notifications.
AI/NLP Assistance Component	To be finalized during implementation	Supporting requirement analysis and skill extraction as an advisory layer to the Requirement Engine (see Section 4.1).
Project Management / Issue Tracking	To be finalized during implementation	Coordinating the development effort itself.

5.2 Development Timeline

The timeline below presents a realistic, week-based academic development schedule for the system described in this SRS. It is organized by activity and deliverable rather than by artificial development phases, and reflects the order in which core modules logically depend on one another (for example, authentication and profiles before requirement submission; requirement submission before matching; matching before booking and team formation).

Week(s)	Activities	Deliverables
1–2	Requirements gathering, stakeholder review of this SRS, finalization of scope and data model.	Approved SRS document; finalized ER model.
3–4	Design of system architecture, UI wireframes, and API contracts for authentication and profile management.	Architecture design document; wireframes.
5–7	Implementation of authentication, account/profile management, and category/service/skill management modules.	Working authentication and profile modules.
8–10	Implementation of requirement submission, requirement analysis integration, and the matching engine.	Functional requirement-to-match pipeline.
11–13	Implementation of individual service booking workflow and worker accept/reject handling.	Functional individual service booking flow.
14–16	Implementation of team formation, Team Leader assignment, project, and task management modules.	Functional team/project/task workflow.
17–18	Implementation of communication and notification modules.	Functional messaging and notification delivery.
19–20	Implementation of payment processing, worker earnings, invoicing, reviews, and reputation calculation.	Functional payment-to-review pipeline.

Week(s)	Activities	Deliverables
21–22	Implementation of verification and complaint/dispute management modules.	Functional verification and complaint workflows.
23–24	Implementation of the administration dashboard (Super Admin, Company Admin, Branch Manager) and reporting/analytics.	Functional administration dashboards.
25–26	System integration testing, security review, and performance testing.	Test reports; defect resolution log.
27–28	User acceptance testing, documentation finalization, and deployment preparation.	UAT sign-off; final documentation set.

6. Appendix

6.1 References

[1] SKAVIA – Main Idea (source project document, provided by the project owner).

[2] SKAVIA – Application/System Idea Document (source project document, provided by the project owner, including detailed workflow and matching-model material).

[3] SRS Template (provided structural template used as the basis for this document's organization).

No external sources, standards, or third-party citations beyond the provided project documents were used in the preparation of this SRS.

6.2 Future Enhancements

The following capabilities are described in the SKAVIA source material as advanced, exploratory, or forward-looking concepts. They are documented here to preserve the project's long-term vision without presenting them as guaranteed current functionality. None of the functional requirements in Section 3.1 depend on these items being implemented.

Future Enhancement	Description
Advanced AI/ML Worker Matching	Replacing or augmenting the current rule-based weighted scoring model with a machine-learning model trained on historical outcomes, once sufficient platform data has been collected.
AI Team Builder	Automated, end-to-end team composition suggested directly by an AI component, beyond the current model in which the Matching Engine ranks candidates and SKAVIA operations personnel confirm the final team.
AI-Based Project Complexity Analysis	Automated estimation of a project's complexity level to guide staffing and scheduling decisions.
AI Price Estimation	Automated suggested pricing for a requirement based on historical data, rather than relying on worker-declared expected rates.
AI Time/Duration Estimation	Automated estimation of how long a service or project is likely to take.
AI Workforce Optimization	Platform-wide optimization of worker allocation across concurrent requirements and branches.
Advanced Demand Prediction	Forecasting of category/service demand to guide worker recruitment and branch planning.
Fraud / Risk Detection	Automated detection of suspicious account, booking, or payment activity.

Future Enhancement	Description
Intelligent Project Monitoring	Automated anomaly detection in project progress (e.g., stalled tasks) beyond manual Team Leader reporting.
Advanced Contract Management	Formal, structured digital contracts between clients and workers/teams beyond the current requirement/project record.
Advanced Invoice Management	Extended invoicing capabilities such as multi-line itemized business invoices and tax handling.
Learning and Training Ecosystem	A structured skill-development or training module for workers.
Skill Assessment and Certification Ecosystem	Platform-native skill testing and certification, as opposed to review of externally issued certificates.
Nationwide Expansion	Extending branch and worker coverage across a wider geographic footprint.
International Expansion	Extending the platform model beyond its initial country of operation.

6.3 Glossary

Term	Definition
SKAVIA	The solution-oriented service and workforce platform described in this document.
Requirement	A structured or natural-language description of a problem or need submitted by a Customer, Business Client, or Organization.
Simple Requirement	A requirement that can be fulfilled by a single worker with a single relevant skill.
Complex Requirement	A requirement that requires multiple skills and, therefore, a team of workers.
Matching Engine	The rule-based component that scores and ranks candidate workers against a structured requirement.
Team Leader	A worker assigned coordination and reporting responsibility for a specific team/project.
Branch	A specialization unit within SKAVIA that focuses on a particular category of services (e.g., SKAVIA Home).
Reputation	An aggregated indicator of a worker's reliability and quality derived from completed work and reviews.
Verification	The process by which SKAVIA confirms a worker's identity, skill, or certificate claims.
Dispute	A formally escalated complaint that requires investigation and a recorded resolution decision.
SKAVIA Service/Management Fee	The portion of the client's total payment retained by SKAVIA for operating the platform and coordinating the service.
Mode	The active context (Client Mode or Worker Mode) that a dual-role user is operating in within a single account.

6.4 Business and Social Value

The SKAVIA source material frames the platform as having intended social and economic value beyond its technical function. These are presented here as intended or potential benefits rather than proven outcomes:

- Creating employment and income opportunities for skilled and semi-skilled workers, including in rural and seasonal contexts.
- Reducing dependence on informal middlemen by connecting clients directly with a coordinated, accountable platform.

- Giving workers improved access to individual customers and businesses beyond their existing personal networks.
- Recognizing and building durable professional reputation for workers through verified reviews and completed work history.
- Supporting broader economic participation by making it easier for small businesses to assemble the mix of skills needed for a project (e.g., a new restaurant) through a single coordinated request.

6.5 Internal Consistency Audit Summary

Before finalizing this document, the following consistency checks were performed, consistent with the audit requirements provided for this SRS:

- Every major feature described in the SKAVIA source material is represented in Section 2.2, Section 3.1, or Section 6.2 (Future Enhancements).
- Every actor identified in Section 2.3 appears in at least one use case diagram (Section 4.2) and at least one detailed use case description (Section 4.3).
- Every functional requirement group in Section 3.1 is represented in the Requirement Traceability Matrix (Section 4.8) with at least one linked use case, module, database entity, and diagram.
- Every entity in the ER diagram (Section 4.4) supports at least one functional requirement and appears in the Data Dictionary (Section 4.5).
- The payment flow (Section 3.1.O), worker earnings (Section 3.1.P), and review/reputation flow (Section 3.1.Q) are sequentially consistent, as shown in Sequence Diagram 5 (Section 4.6.5).
- Team formation (Section 3.1.I), Team Leader responsibilities (Section 3.1.J), and project/task relationships (Section 3.1.K–L) are consistent across the use case model, ER diagram, and Sequence Diagrams 3 and 4.
- Administrative permissions in Section 2.3 are applied consistently in the Administration use case diagram (Section 4.2.3) and in functional requirement group T (Section 3.1.T).
- Business Client is not confused with Company Admin, and Individual Customer is not confused with Business Client, anywhere in this document (see Section 2.3).
- The AI Assistance Layer is consistently presented as advisory input to the Requirement Engine (Section 4.1) and never as a replacement for rule-based matching, payment, or administrative decision-making.
- Future enhancements listed in Section 6.2 are not referenced as dependencies of any functional requirement in Section 3.1.
- No development phases ("Phase 1," "Phase 2," etc.) appear anywhere in this document.
- The confirmed technology stack (Flutter/Dart, Supabase, Supabase Auth, Firebase Storage, Android Studio/VS Code) is applied consistently across Sections 2.4, 2.5, 3.3, 4.1, and 5.1; no unsupported technology, algorithm, or payment gateway has been invented, and every undetermined implementation detail is explicitly marked "to be finalized during implementation."
- The Administration Use Case Diagram (Section 4.2.3) is split into two parts (governance and operational) solely to eliminate excessive line crossings; the underlying actors, use cases, and permissions are identical to those described in Section 2.3 and functional requirement group T.